42

Onconephrology: Kidney Disease and Cancer

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CHAPTER OUTLINE

KEY POINTS

- Acute and chronic kidney disease are common complications in cancer patients and are primarily due to the underlying malignancy or its treatment. Study of the growing connection between the two diseases is known as "onconephrology."
- Acute kidney injury (AKI) is a relatively common complication of cancer and its treatment. Both hematologic malignancies and solid cancers are associated with AKI. The causes can be classified as prerenal (e.g., volume depletion, hypercalcemia, capillary leak, renal sodium wasting), intrarenal (e.g., direct cancer invasion, paraneoplastic effects, drug toxicities), and postrenal (e.g., retroperitoneal cancer or fibrosis, bladder cancer).
- Cancer is frequently complicated by chronic kidney disease (CKD). A number of causes lead to CKD, including the underlying malignancy, drugs used to treat cancer, shared risk factors (e.g., hypertension, diabetes mellitus), nephrectomy for renal cell cancer, and therapies such as hematopoietic stem cell transplantation.
- Diseases involving the glomeruli also complicate various malignancies. Membranous nephropathy, minimal change disease, and proliferative glomerulonephritides are noted paraneoplastic effects of cancer. Paraproteinemias with monoclonal immunoglobulin or light- or heavy-chain synthesis also cause glomerular injury—for example, with AL amyloidosis and monoclonal immunoglobulin deposition disease.
- Electrolyte and acid–base disorders frequently complicate cancer and its therapy. Hyponatremia, hypernatremia, hypokalemia, and hyperkalemia, as well as imbalances in calcium, magnesium, and phosphorus, are the result of direct or paraneoplastic effects or adverse effects of drugs on the gastrointestinal tract (e.g., vomiting, diarrhea) and/or kidneys.
- Hematopoietic stem cell transplantation is a life-saving procedure for patients with certain malignancies. However, this procedure is complicated by AKI, glomerular disease, hypertension, electrolyte abnormalities, and CKD, depending on the patient's underlying risk and the conditioning regimen and type of stem cell transplant used.
- Medications used to treat cancer are associated with adverse renal effects. Conventional chemotherapeutic drugs, immunotherapies, and targeted agents are associated with AKI, glomerulopathies, and electrolyte and acid–base disturbances, and CKD.

Onconephrology refers to the intersection of the disciplines of nephrology and oncology. Kidney disease in multiple forms, ranging from glomerular diseases to electrolyte abnormalities, is common in patients with cancer. In many cases, these kidney disorders are not unique to patients with cancer but can affect the care of the patient, such as affecting the dosage of appropriate chemotherapy in patients with chronic kidney disease (CKD). In other circumstances, either the malignancy itself or its associated therapy may lead to unique kidney diseases, such as acute kidney injury (AKI), glomerulonephritis, or electrolyte disorders. It is the responsibility of the nephrologist to recognize these unique and cancer-associated kidney disorders to provide the most effective therapy.

ACUTE KIDNEY INJURY IN PATIENTS WITH CANCER

EPIDEMIOLOGY

Acute kidney injury (AKI) is a common occurrence in hospitalized patients with cancer and, when it occurs, it is associated with higher costs, increased hospital length of stay, and increased morbidity and mortality. In some cases, AKI may lead to changes in chemotherapy regimens that may lessen the chances of disease remission, or AKI may exclude patients from clinical trials. In the largest study on this topic, 37,267 incident cancer patients in Denmark had follow-up over a 7-year period in the early 2000s.[1] The 1-year risk of AKI, as defined by RIFLE risk (>50% increase in serum creatinine), was 17.5%. The 1-year risk for more severe AKI, the RIFLE injury (>100% increase in serum creatinine), and failure (>200% increase in serum creatinine of >4 mg/dL and requiring dialysis) risk categories was 8.8% and 4.5%, respectively. The 5-year risk for the RIFLE risk, injury, and failure AKI categories was even higher, at 27%, 14.6%, and 7.6%, respectively. Critically ill patients with cancer have a higher incidence of AKI and AKI requiring dialysis than critically ill patients without cancer.[2,3] In those patients with cancer who suffer an episode of AKI, the mortality is elevated, such that the mortality was 13.6% in those without AKI and progressively increased with higher RIFLE stage AKI (risk, 49%; injury, 62.3%; failure, 86.8%).[2] In patients with hematologic malignancies who were undergoing induction therapy, the mortality in those with RIFLE risk AKI was 13.6%, whereas the mortality in those with no AKI was 3.8% over an 8-week period.[4] Importantly, those patients who required dialysis during the period of induction therapy had a mortality of 61.7%.[4] It is important to note that more contemporary studies may demonstrate that outcomes in patients with cancer and AKI are not universally poor, with one study showing that 82% of critically ill cancer patients with AKI completely recovered kidney function, whereas partial recovery was observed in 12%, and chronic renal replacement therapy (RRT) was required in only 6% of patients.[5] The prognosis of AKI greatly depends on the underlying functional and premorbid conditions of the patient, as well as the overall state of the malignancy. Thus, individualized decisions regarding the appropriateness of RRT in these patients is warranted.

A specific group of patients with a high incidence of AKI is those undergoing hematopoietic stem cell transplantation (HSCT). Various studies have shown an incidence of AKI ranging from 23% to 73% according to the different cutoff points used to define AKI.[6–8] Furthermore, the incidence of AKI varies according to the type of HSCT, with most data supporting the finding that myeloablative allogenic transplantation is associated with a higher incidence of AKI.[8]

The risk factors for AKI in patients with cancer are multiple and depend on the specific type of cancer. These can be broadly divided into those that are modifiable and those that are not (Table 42.1). Certain cancers appear to carry a higher AKI risk than others, with renal cell carcinoma, hepatocellular carcinoma, multiple myeloma, and lymphoma having higher rates of AKI.[1] As an example, patients undergoing total (radical) nephrectomy for renal cell carcinoma have an incidence of AKI as high as 33.7%.[9] Even partial nephrectomy to spare nephron loss is associated with AKI, albeit less commonly.[10] Patients with acute lymphoma or leukemia undergoing induction chemotherapy are also at especially high risk to develop AKI, primarily due to tumor lysis syndrome and drug nephrotoxicity. In a series of 537 patients undergoing induction therapy for acute myelogenous leukemia or high-risk myelodysplastic syndrome, 36% developed AKI.[4] Whenever possible, modifiable risk factors should be addressed in an attempt to lower the incidence of AKI.

CAUSES OF ACUTE KIDNEY INJURY IN PATIENTS WITH CANCER

The causes of AKI in patients with solid organ cancers are similar to those of the general population, with an overrepresentation of obstructive nephropathy (especially with

Table 42.1 Risk Factors for Acute Kidney Injury in Patients With Cancer

Modifiable Risks	Nonmodifiable Risks
Nephrotoxic medications (nonchemotherapeutic), such as aminoglycosides, amphotericin B, calcineurin inhibitors	Age > 65 years
Nephrotoxic medications (chemotherapeutic), such as cisplatin, checkpoint inhibitors, ifosfamide, interferon, tyrosine kinase inhibitors, BRAF (serine–threonine protein kinase) inhibitors	Underlying chronic kidney disease, especially diabetic kidney disease
Hypovolemia • True volume depletion—nausea, vomiting, diarrhea • Effective volume depletion—cirrhosis, heart failure, nephrotic syndrome, hypoalbuminemia	Specific cancer types—multiple myeloma, hepatocellular cancer, renal cell carcinoma, pelvic malignancies
Intravenous radiographic contrast	Hematopoietic stem cell transplantation
High tumor bulk and risk for tumor lysis syndrome	Female gender

prostate, ovarian and cervical cancer), chemotherapy-associated nephrotoxicity, sepsis-associated ischemic acute tubular necrosis in the setting of neutropenia, and AKI due to hypercalcemia.[11–13] In patients with hematologic malignancies, the causes of AKI are also similar to those of the general population, with some notable unique causes, such as AKI associated with multiple myeloma (e.g., cast nephropathy, light- or heavy-chain–associated glomerulonephritis or associated with hypercalcemia), tumor lysis syndrome, and tumor infiltration of the kidneys associated with lymphoma and leukemia.[11–13] In those patients receiving a stem cell transplant, there are other situation-specific causes of AKI, including venoocclusive disease and cytokine release syndrome.[14]

Prerenal causes of AKI are common in patients with malignancies and often result from poor oral intake, as well as chemotherapy-induced nausea, vomiting, and diarrhea. Hypercalcemia and its natriuretic effect may also lead to volume depletion and prerenal azotemia. In addition, cancer patients may be prescribed common medications, such as diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, or nonsteroidal antiinflammatory drugs (NSAIDs) that can affect renal autoregulation and exacerbate prerenal AKI. Prerenal causes are so common in this patient population that judicious trials of intravenous (IV) fluids are reasonable in most patients presenting with AKI.

Many of the causes of AKI seen in cancer patients are not unique and are covered in other chapters. However, certain intrarenal and postrenal causes are seen almost exclusively in the setting of malignancy and are covered here and shown in Table 42.2. AKI associated with glomerular disorders, chemotherapeutic agents, HSCT, and hypercalcemia and seen in the setting of renal cell carcinoma are covered in their respective sections in this chapter.

Table 42.2 Intrarenal and Postrenal Causes of Acute Kidney Injury in the Patient With Cancer

Cause	Representative Example
Vascular—thrombotic microangiopathy	Gemcitabine, post–stem cell transplantation
Tubular injury—acute tubular necrosis	Ischemia due to sepsis, nephrotoxicity due to chemotherapeutic agents such as cisplatin
Tubular injury—intratubular precipitation of crystals	Methotrexate
Tubular injury—cast nephropathy	Multiple myeloma
Tubular Injury—lysozymuria	Acute promyelocytic, acute monocytic leukemia Chronic myelomonocytic leukemia
Interstitial injury—interstitial nephritis	Checkpoint inhibitors
Interstitial injury—tumor infiltration of the kidneys	Lymphoma
Glomerular injury	Amyloidosis Rapidly progressive glomerulonephritis Various paraneoplastic or drug-induced glomerulonephritis Paraneoplastic membranous nephropathy
Obstruction	Retroperitoneal lymphadenopathy (lymphoma) Tumor bulk obstructing urine flow (pelvic malignancies)

TUMOR LYSIS SYNDROME

Tumor lysis syndrome (TLS) is encountered in patients with bulky, rapidly growing and chemosensitive malignancies, such as high-grade lymphomas (e.g., Burkitt lymphoma), leukemia, or other cancers with large cellular burdens.[15,16] Typically, TLS is seen after the administration of chemotherapy but can occur spontaneously as well. TLS has been defined by the Cairo-Bishop criteria (Table 42.3) and is biochemically typified by the findings of numerous electrolyte disorders (e.g., hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia) that result from the release of intracellular substances after cell lysis.[15,16] Importantly, TLS may lead to sudden death and seizures due to the electrolyte disturbances encountered in these patients. The mechanism of AKI in these patients is at least partially due to uric acid nephropathy but may also be influenced by cytokine release and nephrocalcinosis (due to high serum calcium phosphate products).[17]

Uric acid nephropathy and AKI are the result of the precipitation of insoluble uric acid in the renal tubules that occurs when purine nucleotides are released from dying cancer cells. These nucleotides are metabolized by xanthine oxidase into uric acid, which is subsequently filtered in the glomerulus and concentrated in the renal tubules. This forms an insoluble precipitate, leading to a combination of intratubular obstruction, vasoconstriction, and inflammation, culminating in a fall of the glomerular filtration rate and AKI.[18] Understanding this pathophysiology leads to rational protocols for the prevention of TLS (Fig. 42.1). The first

Table 42.3 Cairo-Bishop Criteria for Tumor Lysis Syndrome

Type of Criteria	Features
Laboratory criteria for tumor lysis syndrome	Requires two or more of the following criteria within 3 days to or 7 days after initiation of chemotherapy: • Uric acid level ≥ 8 mg/dL or 25% increase from baseline • Potassium level ≥ 6.0 mEq/L or 25% increase from baseline • Phosphorus level ≥ 6.5 mg/dL for children • Phosphorus level ≥ 4.5 mg/dL for adults or 25% increase from baseline • Calcium level ≤ 7 mg/dL or 25% decrease from baseline
Clinical criteria for tumor lysis syndrome	Laboratory tumor lysis syndrome plus one or more of the following criteria: • Creatinine > 1.5 times upper limit of age-adjusted reference range • Cardiac dysrhythmia or sudden death • Seizure

From Cairo MS, Bishop M. Tumor lysis syndrome: new therapeutic strategies and classification. *Br J Hematol.* 2004;127(1):3–11,2004.

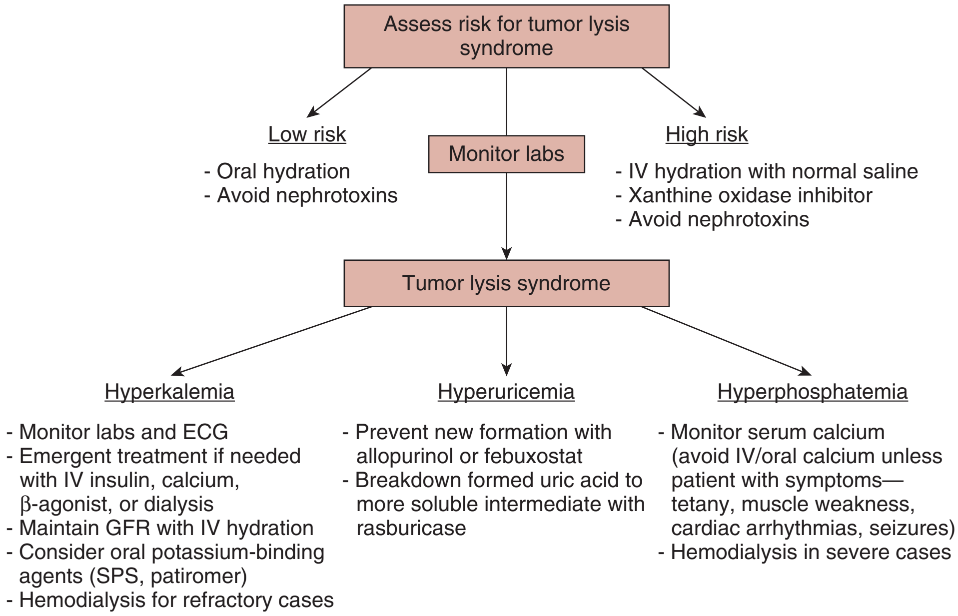

Fig. 42.1 Approach to the management of tumor lysis syndrome. *ECG,* Electrocardiogram; *GFR,* glomerular filtration rate; *IV,* intravenous; *SPS,* sodium polystyrene sulfonate. (From Rosner MH, Perazella MA. Acute kidney injury in patients with cancer. *N Engl J Med.* 2017; 376(18):1770–1781.)

step in a preventive strategy rests on identifying patients at risk for TLS, which should include any patient with a large, rapidly growing and chemosensitive tumor or those patients who, prior to treatment, may already be manifesting signs of tumor lysis with an associated electrolyte disturbances. Low-risk patients should be treated with oral fluids and avoidance of other nephrotoxins; high-risk patients should receive IV 0.9% saline to maintain adequate glomerular filtration and tubular flow rates that facilitate rapid clearance and dilution of uric acid, as well as the clearance of potassium and phosphate. The prophylactic use of a xanthine oxidase inhibitor (allopurinol or febuxostat) is recommended because these drugs will block de novo formation of uric acid and limit further increases in the levels of this compound. For patients who already have elevated uric acid levels, treatment with recombinant urate oxidase (rasburicase) is beneficial for rapidly lowering uric acid levels and is recommended.[19] Rasburicase converts uric acid to a soluble and readily excreted compound, allantoin.[19] It should be cautioned that rasburicase also leads to the production of hydrogen peroxide, which can cause methemoglobinemia and hemolytic anemia in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency, and at-risk patients should be tested for this condition prior to administration.[20]

For those patients manifesting signs of TLS, close monitoring of serum electrolyte levels and clinical status is required (see Fig. 42.1), and electrolyte disorders require rapid therapy. Patients should receive aggressive IV hydration, a xanthine oxidase inhibitor, and rasburicase. For patients presenting with established AKI, emergent hemodialysis to correct electrolyte disorders may be required. Urine alkalinization, which increases uric acid solubility in the renal tubules, is not recommended because it can increase the likelihood of calcium phosphate precipitation.[18]

LYSOZYMURIA

Lysozyme can be released from certain hematologic malignancies (e.g., acute promyelocytic, monocytic, or chronic myelomonocytic leukemia) and is filtered by the glomerulus and then reabsorbed by proximal tubular cells, where it leads to tubular damage and AKI.[21] The condition is rare, and suspected cases can be confirmed by urinary protein electrophoresis, which can identify lysozyme in the urine.

TUMOR INFILTRATION OF THE KIDNEYS

For unclear reasons, leukemia and lymphoma cells have a predilection for infiltrating the kidney and, on autopsy, up to 60% of these patients will have tumor cell infiltrates in the kidney.[22] In most cases, these infiltrates have little clinical significance but, in approximately 1% of cases, they may be so massive that they result in AKI.[22,23] AKI in these cases likely results from compression of the renal parenchyma by tumor cells, with distortion of the microvasculature and tubular architecture. Patients may present with hypertension (HTN), flank pain, and hematuria or, more often, are asymptomatic. Renal imaging with ultrasonography or computed tomography shows bilaterally enlarged kidneys with a heterogeneous texture. A kidney biopsy is diagnostic and reveals tumor cell infiltration. Appropriate chemotherapy with tumor response can lead to rapid improvement in kidney function.

THROMBOTIC MICROANGIOPATHY

Thrombotic microangiopathy (TMA) can be a consequence from the tumor itself or, more likely, from associated chemotherapy, such as gemcitabine or vascular endothelial growth factor (VEGF) inhibitors such as bevacizumab.[24–27] TMA has been associated with mucin-producing gastric, lung, and breast cancers. TMA may also be encountered in patients undergoing HSCT, in the setting of graft-versus-host disease (GVHD, associated with cyclosporine use), and with radiation nephropathy.[27] The presentation of TMA in cancer patients may be subtle and can be acute (after a clearly defined insult) or delayed (months after starting a chemotherapy regimen, such as gemcitabine). In some patients, new-onset and relatively severe HTN and proteinuria may be a clue to the diagnosis of TMA.[26] Unfortunately, the classic laboratory signs of TMA—falling hemoglobin, falling platelet count, and

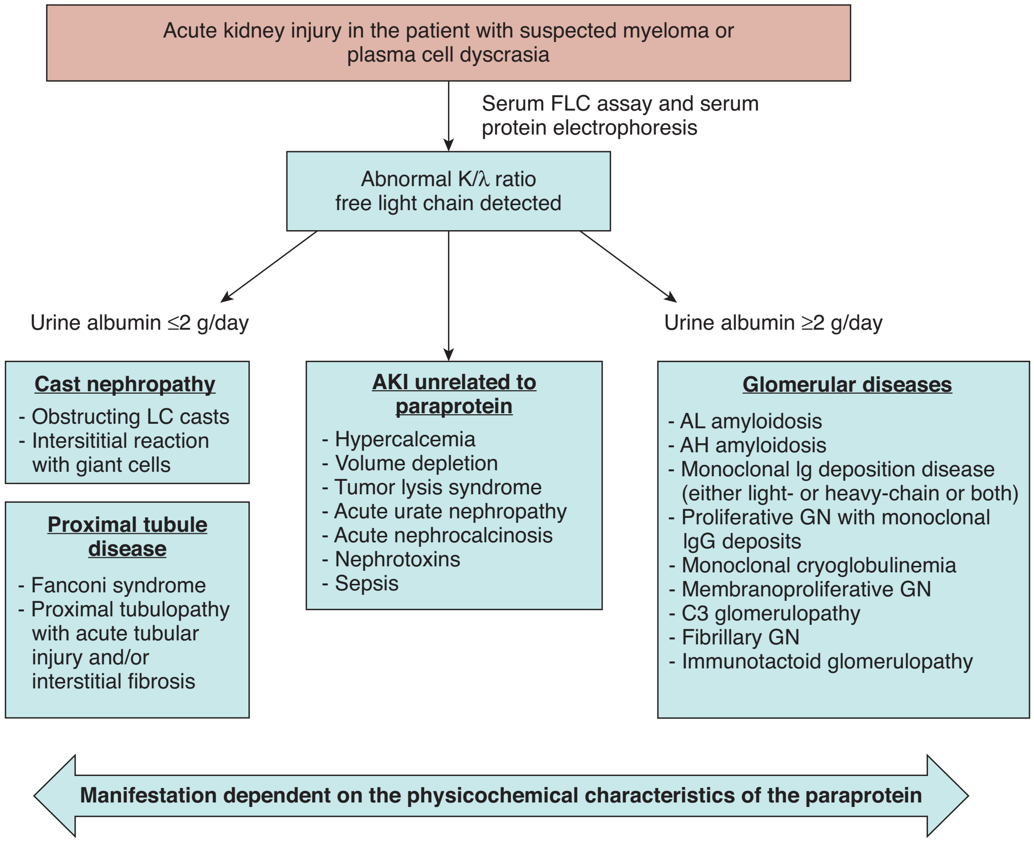

Fig. 42.2 Approach to the patient with acute kidney injury *(AKI)* and myeloma. *FLC,* Free light chains; *GN,* glomerular nephritis; *IgG,* immunoglobulin G; *LC,* light chain.

elevated lactate dehydrogenase level, along with falling haptoglobin and schistocytes on a peripheral blood smear—may not be present or may be mild. In some cases, renal biopsy should be considered for a definitive diagnosis, where pathologic findings of edematous intimal expansion of arteries, fibrinoid necrosis of arterioles, ischemic collapse of capillaries, and focal thrombosis of injured vessels can be seen. There are no trials that guide therapy in these cases, and treatment relies on removing any insults, such as medications, and consideration of using plasmapheresis and rituximab in more refractory cases.[26,27] Malignancy-associated TMA usually responds poorly to plasmapheresis and carries a poor prognosis overall.

LIGHT-CHAIN CAST NEPHROPATHY AND ACUTE KIDNEY INJURY ASSOCIATED WITH MYELOMA

Patients with myeloma have an incidence of AKI that can be as high as 50%.[28] The most common cause of AKI in these patients is due to cast nephropathy, followed by light chain–related proximal tubulopathy, various glomerular diseases (including light-chain deposition disease and AL amyloidosis), hypercalcemia, hyperuricemia, and other contributing factors (Fig. 42.2).

Cast nephropathy is due to the filtration of large amounts of myeloma-produced free light chains into the renal tubules, which bind to Tamm-Horsfall protein (also known as uromodulin) to form insoluble aggregates.[29] These intratubular precipitates lead to intrarenal obstruction and local inflammation, which may be further exacerbated by the proximal tubular uptake of pathogenic light chains, which also leads to activation of inflammatory pathways.[30] Sensitive assays for serum free light chains, serum and urine protein electrophoresis, and immunofixation facilitate the diagnosis of myeloma-related AKI. Typically, patients with myeloma-related glomerular diseases will present with high levels of albuminuria (>2 g/day) indicative of the glomerular injury. Patients with cast nephropathy will demonstrate high levels of urine and serum light chains and lower levels of albuminuria.[31] Of note, 15% of patients with AKI in the setting of myeloma may have a cause of AKI completely unrelated to myeloma; thus, kidney biopsy should be performed in unclear cases.[31]

Therapy for cast nephropathy focuses on a combination of hydration and ensuring good renal tubular flow, correction of hypercalcemia, treatment of other precipitating or aggravating factors, and chemotherapy to lower the myeloma production of pathogenic light chains. Chemotherapeutic regimens for myeloma generally include a proteasome inhibitor, such as bortezomib, which does not require dosing adjustments for renal function.[32] Regimens including bortezomib (usually in combination with dexamethasone and another agent, such as thalidomide or melphalan) have led to renal response rates as high as 72% and dialysis discontinuation in 57% of patients.[33] Other effective agents include thalidomide, lenalidomide, pomalidomide, and carfilzomib.[34] A great deal of controversy remains surrounding the benefit of removing light chains by therapeutic plasma exchange or high-flux hemodialysis; currently, these methods cannot be routinely recommended.[35]

URINARY TRACT OBSTRUCTION

Obstruction of the urinary tract is a common cause of AKI in the patient with cancer. Obstruction can be intratubular due to precipitation of uric acid, as in TLS, or from medications such as high-dose methotrexate or acyclovir.[36] In all cases, knowledge of the risk for intratubular obstruction should lead to the administration of prophylactic IV fluids to maintain urinary dilution of the drug or uric acid and maintenance of tubular fluid flow.

More commonly, urinary tract obstruction will be extrarenal and due to extrinsic compression of the ureters by a tumor mass or due to prostatic disease compressing the urethra. The clinical spectrum of patients presenting with malignant ureteral obstruction was illustrated by a case series of 102 patients.[37] Obstruction was bilateral in 68% of patients, and initial management with a percutaneous nephrostomy or ureteral stent was successful in 95% of cases. Despite successful decompression, 53% of patients developed complications, mostly urinary tract infection and obstruction of nephrostomy tubes or stents, and overall survival was poor (median, 7 months), reflecting the advanced stage of malignancy in such patients. Although renal ultrasonography will usually demonstrate hydronephrosis in patients with obstruction, patients with retroperitoneal fibrosis may not have this finding and, if clinical suspicion is high for this condition, other imaging techniques such as nuclear medicine scans should be pursued.[38]

DECISION MAKING IN THE PATIENT WITH CANCER AND ACUTE KIDNEY INJURY

As discussed earlier, the prognosis of critically ill patients with cancer and AKI is poor, with mortality rates approaching 60% to 70% if RRT is needed. Thus, clinicians may be faced with difficult decisions when offering invasive intensive therapy for such a patient population and may appropriately opt for more conservative and palliative approaches. However, it is important to individualize these decisions because select patients (e.g., those with fewer comorbid conditions, higher functional status, and cancers that are responsive to therapy) may benefit from more aggressive therapy. Thus, decisions regarding the initiation of dialysis therapy require input from the entire care team, as well as the patient and family. These decisions should consider the reversibility of the AKI, long-term cancer prognosis, quality of life prior to critical illness, and the patient's preferences.

CHRONIC KIDNEY DISEASE IN PATIENTS WITH CANCER

EPIDEMIOLOGY

As has been observed with AKI, CKD is also noted to be a possible complication of numerous cancers and their associated therapy. There are a number of explanations for this observation. An important driver of this untoward complication is the fact that preexisting and often unrecognized CKD is common in patients with various types of malignancy. In addition to underlying kidney disease, incomplete renal recovery after multiple episodes of AKI, nephrectomy for kidney cancer with a reduction in nephron mass, and exposure to several courses of nephrotoxic medications can ultimately lead to the development of glomerulosclerosis and tubulointerstitial fibrosis, all of which either exacerbate underlying kidney disease or promote the development of new CKD.

In regard to preexisting CKD in cancer patients, the Renal Insufficiency and Anticancer Medications (IRMA)-1 and -2 studies clearly illustrate the extent of this problem.[39,40] In the two IRMA studies, 52.9% (IRMA-1) and 50.2% (IRMA-2) of patients with an active malignancy had an estimated glomerular filtration rate (eGFR) less than 90 mL/min/1.73 m^2, respectively.[39,40] In this group of patients, the prevalence of stage 3 CKD was 12% (IRMA-1) and 11.8% (IRMA-2), respectively. Stage 4 CKD was fairly rare in these two studies (0.9% and 0.7%, respectively). In an Australian cohort with various cancers (n = 4077) using a prospective population cohort design, eGFR was <60 mL/min/1.73 m^2 in 30% of patients and <45 mL/min/1.73 m^2 in 8.3%.[41] In a cohort of Chinese patients with underlying cancer observed from 2000 to 2004, stage 3 CKD (defined as an eGFR < 60 mL/min/1.73 m^2) was noted in 12.8% (1051 of 8223).[42] The relatively common occurrence of CKD in many cancer patients has been confirmed in other studies, irrespective of the type of malignancy.

OUTCOMES AND PROGNOSIS

Similar to cancer patients who develop AKI, CKD in patients with a malignancy may also suffer from an increased risk for death compared with those without kidney disease.[41–43] Increased cancer patient mortality is not uniformly noted in all studies; however, the effect of CKD on death occurrence appears to be specific to certain cancer types and degree of kidney dysfunction. Cancer patients with stage 3 CKD had an adjusted hazard ratio (HR) for cancer-specific mortality of 1.12 (95% confidence interval [CI], 1.01–1.26).[42] Furthermore, patients with an eGFR less than 30 mL/min/1.73 m^2 had an adjusted HR for cancer-specific mortality of 1.75 (95% CI, 1.32–2.32). As described in a population-based cohort study, stage 3 CKD (eGFR < 60 mL/min/1.73 m^2) was a significant risk factor for death from cancer.[41] In this study, the adjusted HR for cancer-specific mortality was 1.18 (95% CI, 1.08–1.29) for each 10 mL/min/1.73 m^2 decrease in eGFR. It is notable that mortality risk was highest for patients with breast and urinary tract cancers.[41] End-stage renal disease (ESRD) patients on hemodialysis who underwent radiofrequency ablation for hepatocellular cancer (HCC) were identified using the Japanese Diagnosis Procedure Combination database.[44] Four non-ESRD patients for each ESRD patient were matched based on age, gender, treatment hospital, and treatment year. In-hospital mortality was higher in 437 ESRD patients compared with 1345 matched controls (1.1% vs. 0.15%, respectively; odds ratio [OR] = 7.77; P < .001).

In addition to CKD identified by eGFR measurement, the presence of proteinuria also appeared to increase mortality risk in cancer patients. A retrospective study of 9465 patients with newly diagnosed cancer from January to December 2010 observed an increase in all-cause mortality in patients with stage 3 CKD (eGFR < 60 mL/min/1.73 m^2) and proteinuria.[43] As seen in other studies, not all cancers are associated with increased mortality in the presence of CKD. To this point, an eGFR less than 60 mL/min/1.73 m^2 independently predicted death from hematologic (adjusted HR, 2.93; 95% CI, 1.36–6.31) and gynecologic cancers (adjusted HR, 2.82; 95% CI, 1.19–6.70), but not for other cancers. Furthermore, a study in patients with colorectal cancer (n = 3379) noted that stage 3 CKD (eGFR < 60 mL/min/1.73 m^2) was not associated with increased mortality.[45] In addition, a prospective cohort study of cancer patients that examined the risk of death as a secondary endpoint did not show an association between CKD and mortality.[46] In fact, CKD as characterized by a reduced eGFR was not considered a risk factor for death in patients with various types of cancer.[46] Thus, it appears that underlying CKD in cancer patients is associated with increased mortality in some but not all types of malignancy. Clinicians must bear this in mind when counseling patients with cancer about the mortality risks of kidney disease.

CANCER AND CHRONIC KIDNEY DISEASE: BIDIRECTIONAL RELATIONSHIP

As has been previously reviewed, CKD is a well-recognized complication of various malignancies and their therapies. However, in addition to the observed increase in CKD prevalence in cancer patients, both CKD and ESRD appear to be risk factors for the development of a number of malignancies.[47,48] Thus, the relationship between the kidney and cancer can be viewed as bidirectional.[47] The association between severity of kidney disease and the risk of incident cancer was examined over 6,000,420 person-years of follow-up. During this time, 76,809 incident cancers were identified in 72,875 subjects. After adjustment for time-updated confounders, a lower eGFR was associated with an increased risk of renal cancer. The adjusted HR was 2.28 (95% CI, 1.78–2.92) for an eGFR < 30 mL/min/1.73 m^2.[49] The authors also observed an increased risk of urothelial cancer at an eGFR < 30 mL/min/1.73 m^2 but no significant associations between eGFR and other types of cancer. Several observational studies have suggested an increased cancer risk in patients with ESRD on RRT[49–55] (Table 42.4). A recent study from Taiwan has demonstrated that ESRD patients have a significantly higher rate of all cancers (HR, 3.43) with higher specific risks for oral, colorectal, liver, blood, breast, renal, upper urinary tract, and bladder cancer than in the general population.[53] In one study, the observed increased risk for renal parenchymal cancer in ESRD patients was related to the development of acquired renal cystic disease, which increases with time on dialysis.[50] A U.S. study that examined data from 1996 to 2009 observed a higher incidence of cancer in ESRD patients. The cumulative 5-year incidence of any cancer was 9.48% compared with a 5-year cumulative incidence of cancer in U.S. transplant recipients of 4.4%.[54]

Table 42.4 Standardized Incidence Rates of Cancer in Hemodialysis Patients[a]

Type of Cancer	Standardized Incidence Rate
All cancer types	1.42
Kidney parenchyma/pelvis	4.03
Bladder	1.57
Breast (female)	1.42
Non-Hodgkin lymphoma	1.37
Pulmonary, lung	1.28
Colonic, rectal	1.27
Pancreatic	1.08
Prostatic	1.06

[a]See references 49–55.

EFFECT OF CHRONIC KIDNEY DISEASE ON CANCER

The increased mortality observed as a complication in certain cancer patients with CKD compared with those without kidney disease has several plausible explanations. The presence of preexisting CKD may limit the use of effective agents or promote underdosing of anticancer regimens that might otherwise be curative.[47,56] In addition, underlying CKD may alter the bioavailability and/or safety profile of some anticancer drugs, which can potentially lead to different and suboptimal treatment choices. The development of AKI superimposed on CKD, which is a common complication, can lead to the cessation of effective chemotherapeutic regimens, allowing unhindered and potentially rapid tumor growth.[47,56] Furthermore, CKD and ESRD may also be associated with impaired immune surveillance, which would allow cancers to grow and metastasize, especially when combined with insufficient drug dosing. This risk is further increased when these patients receive a kidney transplant, where immunosuppressive regimens (and other factors) enhance the risk for recurrent or de novo cancers that have been associated with increased mortality.[57,58] It is also possible that the adverse renal effects of the anticancer drug used leads to acute and/or progressive kidney injury or to worsening of preexisting CKD, which leads to increased all-cause mortality unrelated to cancer.[47,56] Finally, a pessimistic view of cancer patients with CKD may lead to a nontreatment approach for potentially curable or modifiable malignancies.

ASSESSMENT OF KIDNEY FUNCTION

Patients with cancer require frequent assessment of kidney function to ensure proper dosing of chemotherapeutic agents.[47,56,59] In addition, close monitoring during ongoing anticancer therapy is critical for timely recognition of drug-induced nephrotoxicity (either AKI or progression of underlying CKD).[56,59] The Kidney Disease: Improving Global Outcomes (KDIGO) guidelines have recommended that the glomerular filtration rate (GFR) should be the standard measure to evaluate kidney function for CKD staging and drug dosing purposes.[60] In addition, the guidelines also state that clinicians should use the most accurate method to assess kidney function in the individual cancer patient. This is of particular importance because the incidence of CKD in patients with cancer is as high as 50%.[39] Lack of appreciation for the reduced GFR in patients with CKD can lead to erroneous, harmful, and potentially life-threatening dosing of medications (and their metabolites) that are cleared predominantly by the kidneys. However, the appropriate methodology for assessing GFR in cancer patients is currently unknown. Several small studies have examined the accuracy and precision of traditional eGFR formulas in cancer patients, including the four-variable

Modification of Diet in Renal Disease (MDRD), Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI), and two formulas derived specifically in cancer patients (the Wright and Martin formulas).[61,62] The previously described bedside GFR formulas all performed similarly well when compared with GFR determined by technetium-99m diethylenetriamine pentaacetic acid (^{Tc99m}DTPA) clearance and when used for selecting the appropriate drug dosage for carboplatin.[63] It is worth noting that all formulas significantly underdosed carboplatin, highlighting the need for continued research to determine the most accurate methodology for assessing GFR in this growing patient population. Currently, we recommend use of the CKD-EPI formula for determination of eGFR in most patients with cancer. Note that the eGFR is not reliable in the setting of AKI, and the KDIGO guidelines have therefore recommended that timed clearances of urea and creatinine may be of great value for persons with AKI.[60] See Chapter 23 for detailed discussion of methods for measuring and estimating GFR.

DRUG DOSING IN CHRONIC KIDNEY DISEASE AND END-STAGE RENAL DISEASE

Anticancer drugs are cleared from the body predominantly by the kidneys, extrarenal pathways, or a combination of the two excretory pathways. One of the most important drug-related problems in cancer patients with impaired kidney function is inappropriate medication use and dosing errors. For example, many cytotoxic drugs and their active and potentially toxic metabolites are eliminated by the kidneys via some combination of glomerular filtration, tubular secretion, and/or tubular reabsorption. Furthermore, CKD patients receiving chemotherapeutic agents often have alterations in one or more of their pharmacokinetic parameters.[64] Alterations in medication absorption, volume of distribution and drug distribution in the body, protein binding, biotransformation, and kidney excretion may result in the accumulation of potentially toxic components and overdosage.[65,66] It is therefore critical that clinicians be wary to adjust doses of drugs that are excreted primarily by the kidneys appropriately. Drug dosing will use either the calculated or measured creatinine clearance or estimated GFR formulas (as discussed earlier), which will allow for safer use of chemotherapy in cancer patients with underlying CKD. Drug dosing and appropriate adjustments for the most commonly used chemotherapeutic agents, which may require dose modification in the setting of underlying kidney disease, are noted in Table 42.5.[63,67,68]

It is important to recognize that CKD also alters the pharmacokinetics of drugs that are cleared by nonrenal pathways. It has been demonstrated that uremia (both AKI and CKD) affects hepatic drug metabolism and coupled transport.[69] It has been shown that uremic toxins interfere with transcriptional activation, cause downregulation of gene expression mediated by proinflammatory cytokines, and directly inhibit the activity of the cytochrome P450 enzymes and drug transporters.[70] These additional nuances in pharmacokinetics will make prescribing of some drugs to CKD patients complicated, and developing a rational strategy for drug dosing in the CKD population can be quite challenging for clinicians. As a result, clinicians will need to watch closely for symptoms and signs of drug-related overdose and toxicity when the patient otherwise has had appropriate renal dosing of the anticancer drug regimen. Thus, clinicians should work closely with clinical pharmacists when these drugs are prescribed to CKD patients, and special considerations should be taken to optimize exposure to cytotoxic drugs and reduce the risk of adverse effects.[71] Precautions should include consideration of current medication pharmacokinetic and pharmacodynamic information, use of less nephrotoxic agents, and documented preventive measures, which are well integrated into oncology practice and vigilant surveillance.[72,73]

The management of cancer in the ESRD population is particularly challenging. Very little is known about anticancer drug management in ESRD patients and even less about the optimal timing and necessary dosage adjustments, depending on dialysis. In ESRD patients on RRT, kidney function no longer contributes a significant amount of drug clearance. As a result, these patients may require a reduction in drug dose to avoid excessive drug exposure and associated toxicity. This is clearly needed with drugs that are normally eliminated by the kidneys and also when extrarenal (hepatic primarily) metabolic and excretory pathways are impaired due to uremia. Because most cytotoxic drugs used are excreted predominantly in the urine as unchanged drug or active or toxic metabolite(s), any reduction in kidney clearance (as in CKD and ESRD) may result in the accumulation of potentially toxic components and overdosage.[74] The dosage of chemotherapeutic agents used in these patients will thus frequently require dosage reduction to avoid severe toxic effects. Careful dosage adjustment is mandatory to optimize exposure to cytotoxic drugs and reduce the risk of adverse effects.

The efficiency (or lack thereof) of drug clearance by the modality of dialysis must also be considered to allow for appropriate timing of chemotherapy administration. Thus, it is necessary to determine what fraction of drug(s) is removed by hemodialysis to allow chemotherapy dosing after HD sessions (medications with excellent dialytic clearance) to avoid drug removal, which may result in a loss of anticancer efficacy. Drugs that are not removed efficiently by dialysis can be administered at any time, either before or after the dialysis session. Hemodialysis influences drug pharmacokinetics primarily by the following three properties—drug clearance by dialysis, extraction coefficient, and dialysis extraction factor.[74] Hemodialysis clearance is the removal rate relative to blood concentration when entering the dialyzer. It is calculated by the amount removed (in mg/min) relative to the rate of presentation (mg/mL). Extraction coefficient, also called the "extraction ratio," is the percentage of drug removed from blood across the dialyzer. It is calculated by the rate of removal (mL/min) relative to the rate of presentation (mL/min).[75] Dialysis clearance and the extraction coefficient measure the ability of a dialysis system to remove a drug from the blood, but do not indicate how readily the drug is removed from the body. Because hemodialysis clearance and extraction ratio cannot be extrapolated to the clinical setting, the dialysis extraction factor of the drug (when available) should be used to determine the clinical impact of dialysis session on drug removal. This extraction factor, which is derived from total body clearance and the dialysis clearance of a drug, represents the actual influence of dialysis on drug pharmacokinetics. Hemodialysis is considered clinically relevant if the dialysis extraction factor exceeds 25% of the overall drug elimination.[76,77]

Table 42.5 Dose Modifications for Commonly Used Chemotherapeutic Agents in Chronic Kidney Disease Patients[63,67,68]

	Creatinine Clearance (mL/min)			
Agent	**90–60**	**60–30**	**30–15**	**Hemodialysis**
Bleomycin	100%	75%	75%	50%[a]
Capecitabine	100%	75%	Avoid	Avoid
Carboplatin	Dosing based on AUC Calvert formula—total carboplatin dose (mg) = target AUC × (eGFR + 25); AUC varies between 5 (treated pts) and 7 (untreated pts; mg/mL × min)			In dialysis patients, consider GFR = 0 and target dose = 125–175 mg; *E*, 84% ± 3%[b]
Carmustine	100%	80% if CrCl ≥ 45; 75% if CrCl < 45	Consider alternative	Consider alternative
Cisplatin	100%	50%	50%	25%–50%[b]
Crizotinib	100%	100%	50%	NA
Cyclophosphamide	100%	75%–100%	75%	75%[a]; *E*, 40%–90%
Cytarabine (high dose)	100%	60% if CrCl ≥ 45; 50% if CrCl < 45	30% or consider alternative	30% consider alternative[a]
Dacarbazine	100%	80%–70%	NA	100 mg daily for 5 days per cycle
Eribulin	100%	≥40 mL/min: 100%; <40 mL/min—NA	NA	NA
Etoposide	100%	75%	75%	50%[c]
Fludarabine	100%	80%, United States; 50%, United Kingdom	Avoid	Avoid
Hydroxyurea	100%	50%	50%	20%[b]
Ifosfamide	100%	75%–100%	75%–100%	75%[b]; *E*, 87%
Irinotecan	100%	NA	NA	30-40%
Lenalidomide	25 mg daily	10 mg daily	15 mg every 2 days	15 mg every 2 days; *E*, 31%[b]
Lomustine	100%	75%	75%	25%–50%
Melphalan	100%	75%	75%	50%
Methotrexate	100%	80%	50%	Avoid; HD Cl, 92 ± 10 mL/min[b]
Mitomycin	100%	100%	100%	75%[a]
Oxaliplatin	100%	100%	Avoid if CrCl < 30 (Canada) or < 20 (United States)	Avoid
Pentazocine	100%	66%	66%	50%
Pentostatin	100%	50%–75%	50%	50% (1–2 h before dialysis)
Pemetrexed	100%	100%	Avoid	Avoid
Regorafenib	100%	100%	NA	NA
Sorafenib	100%	100%	100%	25% of the dose; increase to 100% according to clinical safety and efficacy[c]
Sunitinib	100%	100%	100%	25%; then increase to 100%, according to clinical safety and efficacy; *E*, 0[c]
Topotecan	100%	50%	50%	25%[b]
Vandetanib	100%	25%	NA	NA

[a]In the absence of data on its removal during dialysis, the drug will be administered after the session.
[b]The drug is dialyzable. Therefore, it will be administered after the session, on days with HD.
[c]The drug may be administered indifferently, before or after the HD session.
AUC, Area under the curve; *E,* extraction coefficient (%); *NA,* not available; *pts,* patients.

Ultimately, it is very important to measure the pharmacokinetic parameters in ESRD patients on hemodialysis and to compare these data with those of patients with normal kidney function. The current lack of pharmacokinetic information of anticancer drug therapy in ESRD patients underlines the need for prospective studies in these patients.

CANCER SCREENING IN END-STAGE RENAL DISEASE

Although cancer screening recommendations for the general population focus on early detection and treatment to improve outcomes and survival,[78] it is not clear that these same principles apply to ESRD patients. This group of patients has many unique differences from the general population that may make screening more or less relevant.[79] These include an attenuated lifespan, occurrence rates (common vs. rare) of specific cancers, unknown accuracy of screening tools, and efficacy of interventions when a cancer is identified.

The adjusted survival for incident hemodialysis and peritoneal dialysis patients in 2008 was only 55% and 66% at 3 years after ESRD onset, respectively, illustrating the marked lifespan difference as compared with the general population.[80]

ESRD patients younger than 80 years of age live less than one-third as long as their non-ESRD counterparts, whereas those aged older than 80 years are projected to live less than half as long as those without ESRD.[80] The most common cause for mortality, accounting for approximately 50% of deaths, is cardiovascular disease; malignant deaths are relatively uncommon.[80] Importantly, individual characteristics should be taken into account because groups include those with higher and lower mortality risks. For example, the 50-year-old man with uncontrolled type 2 diabetes mellitus, HTN, heart failure, and peripheral vascular disease has a much higher mortality than the 50-year-old man with polycystic kidney disease and no other significant comorbid conditions. Thus, a patient-specific understanding of the expected lifetime is required but is not routinely available for decision making. However, it is fair to conclude that ESRD patients have a shorter lifespan than the general population, and it is not clear that a uniform approach to cancer screening adds significant length or quality of life.

The goal of cancer screening is early cancer detection to enhance patient longevity. This appears to be an effective strategy for many cancers (e.g., using flexible sigmoidoscopy for colon cancer screening) in the general population when a patient's expected lifespan exceeds 10 years.[81] Thus, colon cancer screening is unlikely to be effective in many ESRD patients with multiple comorbidities whose lifespan is 3 to 5 years. In addition, screening requires patients to undergo additional procedures and tests, which have the potential for harm. Thus, in the absence of a survival benefit or if the survival benefit of screening is only realized in those with an expected lifespan of more than 10 years, cancer screening in the general ESRD population is unlikely to improve outcomes or be cost-effective. It is noteworthy, however, that certain ESRD patients enjoy life expectancies more than 10 years and may benefit from individualized cancer screening. To make this recommendation, it is important to understand the incidence of various malignancies in this population. Both CKD and ESRD are associated with an increased incidence of certain cancers,[49–55] some of which may be amenable to screening strategies.

SCREENING TESTS FOR CANCER IN THE END-STAGE RENAL DISEASE PATIENT

Cancer screening in ESRD patients is also highly dependent on the sensitivity and specificity of screening tests used in the general population. Unfortunately, studies examining the accuracy of screening tests are generally performed on highly selected patient populations with minimal comorbidities.[82] Thus, the extrapolation of these test findings to those with ESRD is highly questionable. For most commonly used screening tests, large-scale clinical studies assessing their predictive value in detecting cancer have not been performed. In addition, there are a number of important caveats for several screening tests in the ESRD population. False-positive screening tests are observed in ESRD patients due to underlying issues such as uremic platelets (positive fecal occult blood), mammary microcalcifications, dense breast tissue (positive mammography), and elevations in several tumor markers in the absence of cancer.[83] Importantly, false-positive screening tests may lead to further inappropriate diagnostic tests and workups, unnecessary costs, and minimal benefits.

OUTCOMES OF CANCER THERAPY IN END-STAGE RENAL DISEASE PATIENTS

In view of the increased incidence of cancer in ESRD patients, the most important question is whether a malignancy identified by screening is amenable to effective and safe therapy. Unfortunately, robust data on the outcomes of cancer therapy in patients with underlying ESRD are not available. Although cancer in this group is generally treated in the same manner as in the general population, there are important differences due to difference in pharmacokinetics and pharmacodynamics in ESRD (previously reviewed). In ESRD patients, underlying drug dosing modifications and the challenges related to the highly variable dialyzability of drugs limit the applicability of outcome data generated in the general population.[84] Cancer response rates may be very different with adjusted dosing and dialytic removal, which likely affect drug levels and overall drug exposure.

Although efficacy concerns abound for anticancer drug therapy, adverse drug effects are also higher in ESRD patients and can lead to increased morbidity and mortality. Clearly, more data are required on the efficacy of chemotherapeutic regimens in ESRD patients, and an individualized approach, which includes nephrologists, pharmacists, and oncologists, is needed to maximize outcomes and minimize the complications of drug therapy. In addition to drug toxicity, studies have observed increased adverse outcomes in patients with ESRD undergoing surgical treatment for lung and colorectal cancers.[85–87] ESRD patients with colorectal cancer undergoing tumor resection suffered a higher risk of re-intubation and a need for longer ventilator support, as well as higher rates of sepsis, deep surgical site infections, and pneumonia.[85] Most importantly, the patients also had a much higher risk of death.[85] Similar findings were found in ESRD patients undergoing surgical resection for lung cancer.[86,87] These concerns question the benefit of screening when outcomes with treatment appear to be suboptimal.

RECOMMENDATIONS FOR CANCER SCREENING IN END-STAGE RENAL DISEASE PATIENTS

Data from routine cancer screening studies in ESRD patients have not shown this approach significantly increases life expectancy or is cost-effective.[88–91] For example, the life expectancy increase for a 60-year-old women with ESRD and type 2 diabetes mellitus who has undergone breast cancer screening would range from 1 to 16 days.[88] Furthermore, a cost analysis has demonstrated that the costs per unit of survival benefit from cancer screening range from 1.6 to 19.3 times higher in ESRD patients, with the gain in life expectancy only 5 days or less.[90] A Canadian study investigating the utility of breast and cervical cancer screening in ESRD patients has demonstrated similar results.[89] Although these points are well taken, most of these studies were performed more than 10 years ago, and mortality improvements have been seen in the ESRD population, which changes the life expectancy calculations. Also, individual patients with unique cancer risks and longer life expectancies are not usually included in these analyses. A cohort study in ESRD patients aged older than 50 years has noted that colon cancer screening is performed more often among healthier patients and those with the highest likelihood of transplantation; the overall screening rate was eight times higher than the rate in non-ESRD patients with similar limited life expectancies.[92]

Despite the fact that the general ESRD population may not benefit from universal cancer screening, an individualized approach is critical to identify patients who might benefit. The patient should be assessed for the following: (1) life expectancy; (2) risk of cancer; (3) transplantation candidacy; and (4) his or her perspective and preferences. Gauging patients prognosis and life expectancy help in the decision making process. A poor prognosis is likely when two or more of the following factors are present: (1) age older than 75 years; (2) high comorbidity score on a tool such as the Charlson score or a negative response to the surprise question (i.e., "Would I be surprised if this patient was alive in 1 year?"); and (3) poor functional status or poor nutritional state.[93] A number of prognostic scoring tools can be used to estimate life expectancy and prognosis.[93–96] In contrast, patients with an acceptable life expectancy may engender benefit from age-appropriate cancer screening, as do patients with a strong family history of cancer or another specific cancer risk (e.g., acquired cystic kidney disease). In young patients and those considered transplantation candidates, screening for acquired cystic kidney disease is appropriate. Cancer screening is reasonable for patients who are transplantation candidates determined to have an acceptable longevity. Finally, patient preferences are important, and a discussion of the risks, benefits, and costs of cancer screening is warranted.

GLOMERULAR DISEASES ASSOCIATED WITH CANCER

Patients with cancer can present with a host of glomerular diseases either related to the primary cancer or to its therapy. Glomerular diseases associated with chemotherapy are discussed in the section on chemotherapeutic agents. Glomerular diseases associated with HSCT are discussed in the section on HSCT. Typically, patients present with hematuria, proteinuria, nephrotic syndrome, and variable degrees of renal insufficiency. For those patients with a known malignancy who develop proteinuria and/or hematuria, a malignancy-related glomerular disease should be suspected. Table 42.6 lists the commonly seen glomerular diseases associated with malignancies and their treatment. Many of these associations are based on isolated cases and case series and are likely rare associations; however, they should be considered in patients with the right presenting symptoms. Some glomerular disorders in patients with cancer represent paraneoplastic syndromes in which the clinical manifestation is not directly related to the tumor burden, invasion, or metastases but instead is caused by the action of a secreted tumor product, such as a cytokine, growth factor, or tumor cell antigen.[97] There is little information about the prevalence of paraneoplastic glomerular diseases.[98] For example, in one study of 120 patients with glomerulonephritis, 14.1% had a concomitant cancer but causality could not be determined.[99] Furthermore, it is important to realize that the treatment of certain glomerular diseases with immunosuppressive therapies (e.g., cyclophosphamide) can increase the risk for subsequent cancers.

The diagnosis of paraneoplastic glomerular diseases is difficult, and determining causation between the glomerular lesion and the cancer can be facilitated by carefully reviewing the sequence of clinical events and the timing of the cancer diagnosis in relation to the diagnosis of the glomerular lesion. In one study, most cancer diagnoses that were thought to be related to the glomerular lesion were either found at the time of renal biopsy or within the first year afterward.[99] The importance of determining the link between the cancer and glomerular disease is that if this link is present, treatment of the cancer may result in remission of the glomerular lesion.

Table 42.6 Glomerular Diseases Associated With Cancer and Chemotherapeutic Agents

Glomerular Lesion	Malignancy	Drug
Membranoproliferative glomerulonephritis	Lung, renal cell, breast, esophageal, gastric, Wilms, melanoma, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia, monoclonal gammopathy, myeloma	Gemcitabine, sirolimus
Minimal change disease	Lung, colon, pancreas, bladder, renal cell, ovarian, mesothelioma, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia	Interferon-α, β, γ; pamidronate, doxorubicin, daunorubicin, sirolimus
Focal segmental glomerulosclerosis (FSGS)	Lung, renal cell, breast, esophageal, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia, T cell leukemia	Sirolimus, temsirolimus, everolimus, doxorubicin, daunorubicin, bisphosphonates
Collapsing FSGS	Unlikely, but causes of FSGS listed above could be considered	Interferon-α, β, γ; pamidronate, gefitinib, sirolimus, doxorubicin, daunorubicin, clofarabine, bisphosphonates
Membranous nephropathy	Lung, colon, gastric, pancreas, prostate, breast, head and neck, Wilms, teratoma, ovarian, cervical, endometrial, melanoma, skin, pheochromocytoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia	Sirolimus
Lupus-like glomerulonephritis	None	Ipilimumab
IgA nephropathy	Lung, pancreas, renal cell, head and neck, Hodgkin and non-Hodgkin lymphoma	Sirolimus

MEMBRANOUS NEPHROPATHY

Membranous nephropathy is the most commonly reported glomerular disease in patients with cancer.[100–103] The database for this assertion is small and, in the largest study of 240 patients with membranous nephropathy, 10% of patients were found to have cancer that was more commonly found in the lung, prostate, and stomach.[102] In this case series, 21 cases of cancer were identified at the time of renal biopsy, and the remaining 3 cases were identified within 1 year postbiopsy. Overall, the standardized incidence ratio of cancer in patients with membranous nephropathy is 2.25, with 80% of the cancers detected before or at the time of renal biopsy.[104] From case-series data, the cancers that have been most often associated with membranous nephropathy include lung, gastric, renal cell, and prostate cancers and thymomas.[100,101] Other cancers that have been reported with membranous nephropathy include breast, colorectal, pancreatic, esophageal, and hepatic cancers.[100,101] In those patients who are aged older than 65 years or have a significant smoking history, the incidence of malignancy with membranous nephropathy may be higher.[102]

Mechanistically, malignancy-associated membranous nephropathy may be due to one of several mechanisms that lead to subepithelial deposits forming in the glomerular basement membrane.[104] Tumor cells may shed antigens that can do the following: (1) form circulating immune complexes that become trapped in the capillary wall and elicit an immune reaction in a subepithelial location; (2) form immune complexes that initially deposit in a subendothelial location but dissociate and reform in a subepithelial location; and (3) deposit in a subepithelial location (depending on the size and charge of the antigen) where they can react with a circulating antibody. A fourth mechanism may be related to an infection with an oncogenic virus or altered immune function, which can lead to both malignancy and membranous nephropathy.

Distinguishing primary versus secondary (malignancy-related) membranous nephropathy is challenging but perhaps made easier by the recognition that a high percentage of primary membranous patients have an elevated circulating level of autoantibodies to the podocyte transmembrane glycoprotein M-type phospholipase A2 receptor (anti-PLA2R) or increased expression of anti-PLA2R on kidney biopsy.[105–107] Table 42.7 lists features of malignancy- associated membranous nephropathy that can be helpful in diagnosis. There should be a high index of suspicion for an underlying malignancy in patients with a new diagnosis of membranous nephropathy, especially if there are other risk factors for cancer. However, the most efficient and cost-effective method for screening these patients has not been identified.[108] It is recommended that age- and gender-appropriate cancer screening be performed on patients with a new diagnosis of membranous nephropathy. In those patients in whom no malignancy is found on initial screening, follow-up vigilance for possible cancer should be maintained because there are case reports of cancers being detected up to 5 years after the initial diagnosis of membranous nephropathy.[109] Whether these cancers are directly related to the occurrence of glomerular disease is not known.

Malignancy-associated membranous nephropathy should resolve with surgical removal of the tumor or with effective chemotherapy.[104] In the absence of curative therapy, standard therapy for membranous nephropathy can be attempted but may have a lower likelihood of inducing remission.

Table 42.7 Factors That Distinguish Primary From Malignancy-Associated Membranous Nephropathy

Parameter	Solid Tumor–Associated Membranous Nephropathy
Clinical clues	Age > 65 years, smoking > 20 pack-years, family history of malignancy, history of thromboembolism
Serologic	Absence of circulating anti-PLA2R autoantibodies in serum.
Histopathologic, kidney biopsy findings	1. Predominance of glomerular IgG1, IgG2 deposition (idiopathic membranous nephropathy is associated with IgG4 predominance) 2. Normal glomerular anti-PLA2R staining 3. Presence of > 8 inflammatory cells/ glomerulus

OTHER GLOMERULAR DISEASES ASSOCIATED WITH SOLID TUMORS

There are associations between solid-organ malignancies and glomerular diseases for minimal change disease (e.g., lung, colorectal, thymoma, and renal cell cancers); focal segmental glomerulosclerosis (e.g., renal cell carcinoma, thymoma); membranoproliferative glomerulonephritis (e.g., lung, renal cell, and gastric carcinomas); IgA nephropathy (e.g., renal cell carcinoma, upper respiratory tract tumors, nasopharyngeal cancers), Henoch-Schönlein purpura (e.g., lung, upper respiratory tract tumors, digestive tract tumors); crescentic glomerulonephritis (e.g., renal cell, gastric, and lung carcinomas) and AA-type amyloidosis (e.g., renal cell carcinoma).[104] These potential associations highlight the fact that in patients presenting with glomerular diseases, clinicians should remain vigilant for the possibilities of an underlying associated malignancy. Age- and gender-appropriate cancer screening should be undertaken in these patients. The association is made stronger if effective therapy for the cancer leads to concomitant remission of the glomerular disease.

HEMATOLOGIC MALIGNANCY–ASSOCIATED GLOMERULAR DISEASES

Both lymphomas and leukemias have been associated with an increased risk of various forms of glomerular diseases.[110] The associations are strongest between Hodgkin disease and minimal change disease and chronic lymphocytic leukemia (CLL) and membranoproliferative glomerulonephritis (MPGN).

HODGKIN DISEASE

Minimal change disease is the most frequent glomerular disease seen in association with Hodgkin disease but is uncommon, with an incidence of 0.4% in two large studies.[111] Typically, the nephrotic syndrome occurs early in the disease, with up to 40% of cases presenting at the time of or just prior to diagnosis of the cancer.[112] Minimal change disease in this setting is typically both steroid- and cyclosporine-resistant, and effective therapy of the lymphoma usually induces remission of the nephrotic syndrome.[104] Importantly, recurrence of the nephrotic syndrome in patients with previously treated Hodgkin disease can signal the recurrence of lymphoma.[111,112] The pathogenesis of minimal change disease in Hodgkin disease may be related to the induction of *c-maf*–inducing protein (c-mif), which can lead to cytoskeletal disorganization and foot process effacement of the podocyte.[113]

CHRONIC LYMPHOCYTIC LEUKEMIA AND B-CELL LYMPHOMAS

The nephrotic syndrome is seen in approximately 1% to 2% of patients with CLL.[114] Typically, the most common glomerular lesion seen in patients with CLL is MPGN, followed by membranous nephropathy.[115] The pathogenesis of MPGN in these cases may involve cryoglobulin production by the abnormal B cell clone. In addition, B-cell lymphomas may produce a pathogenic immunoglobulin that can lead to a host of deposition diseases, including monoclonal immunoglobulin deposition disease (MIDD) and immunotactoid glomerulopathy, in which the deposits are organized in a microtubular array.[104] A clear link between the lymphoma and glomerular disease is solidified when effective chemotherapy leads to a remission of both the lymphoma or CLL and nephrotic syndrome.

PLASMA CELL DYSCRASIA

Plasma cell dyscrasias are associated with a host of glomerular diseases and nephrotic syndrome, the most common of which is AL amyloidosis; this is found in up to 5% to 11% of patients with myeloma at autopsy.[116] Other glomerular diseases include light-chain deposition disease (up to 50% of these patients are found to have myeloma) and heavy-chain deposition disease (up to 25% of these patients have myeloma).[117] In AL amyloidosis, the predominant light-chain isotope is lambda (λ), whereas in light-chain deposition disease, it is kappa (κ).[104] The diagnosis of glomerular diseases in a patient with suspected myeloma is facilitated by finding urine albumin excretion rates > 2 g/day, and a renal biopsy is indicated in these patients (see Fig. 42.2). Several other rarer forms of glomerular disease can be found in the presence of a plasma cell dyscrasia and paraproteinemia. In all cases, effective chemotherapy or HSCT can lead to remission of the glomerular disease.[104]

FLUID AND ELECTROLYTE ABNORMALITIES ASSOCIATED WITH MALIGNANCIES

It is not uncommon to encounter electrolyte imbalances in patients with malignancies. In general, these abnormalities may be directly linked to the malignancy as a paraneoplastic process or as a result of chemotherapeutic regimens. Conversely, they may be unrelated to the malignancy and may be similar to electrolyte disorders seen in patients without cancer (e.g., diuretic-induced hypokalemia).

HYPONATREMIA

Disorders of water homeostasis leading to hyponatremia are among the most common electrolyte disorders encountered in patients with cancer, with a wide range of prevalence reported in the literature (4%–47%).[118,119] In patients presenting to the hospital with hyponatremia, approximately 14% had an underlying malignancy.[120] Perhaps most importantly, hyponatremia in cancer patients is associated with poor outcomes; the hospital length of stay is doubled, and mortality (within 90 days) is increased three to fivefold.[118,121–123] Whether hyponatremia has a direct link to these poor outcomes is debatable; it is more likely that hyponatremia is a marker of severity of illness, disease progression, and overall debility.

The symptoms attributable to hyponatremia are generally nonspecific (e.g., lethargy, nausea) or neurologic (e.g., headache, confusion, seizures) in origin, and the diagnosis relies on correlating laboratory studies with these symptoms. Elucidating the cause of hyponatremia requires careful history taking, physical examination, and judicious laboratory evaluation. An approach to the patient with hyponatremia and cancer is shown in Fig. 42.3. Clearly understanding the cause of a patient's hyponatremia is critical in tailoring appropriate therapy.

The syndrome of inappropriate antidiuretic hormone secretion (SIADH) is a common cause of hyponatremia in the patient with cancer and has numerous causes, ranging from paraneoplastic to medication-induced.[124] Paraneoplastic causes include small cell lung cancer (most common and seen in 10%–15% of patients) and head and neck tumors. Case reports have described SIADH with numerous other solid-organ malignancies, from brain to sarcoma.[124–126] Hyponatremia in these patients typically develops slowly and insidiously, and patients rarely have symptoms unless their serum sodium level is <125 mmol/L.[127] Serial measurements of ADH, at least for small cell lung cancer, reflect the state of the cancer, and levels fall with remission and rise with recurrence.[128] Many chemotherapeutic agents can also lead to SIADH. These include cyclophosphamide, cisplatin, vinblastine, and vincristine.[129] Many of the associations between chemotherapeutic agents and hyponatremia are derived from case reports, and the causal linkage between the drug and SIADH is not conclusive. Of note, in many cases, chemotherapeutic agents (including cyclophosphamide) are administered with aggressive IV hydration, which, depending on the levels of ADH and urine osmolality, may lower serum sodium levels further.

Therapies for hyponatremia in the patient with cancer are similar to those for other causes of hyponatremia and should be guided by three factors: (1) the presence of hyponatremia-related symptoms; (2) the duration of hyponatremia; and (3) the volume status of the patient. When possible, removal or correction of the underlying cause should always be pursued. In the vast majority of cases, hyponatremia associated with cancer is chronic in nature; thus, correction rates should be slow (no more than 6 to 8 mmol/L/day). In the presence of severe neurologic symptoms (e.g., seizures, obtundation, coma), 3% hypertonic saline can be administered as small boluses or continuous infusion to raise the serum sodium level by 4 to 6 mmol/L, which usually improves these symptoms dramatically.[130] Fluid restriction (typically defined as oral intake that is 500 mL less than urine output) is a reasonable treatment for patients with mild and transient hyponatremia due to SIADH. However, given the need for hydration protocols with several chemotherapies, as well as the need to maintain adequate nutrition, fluid restriction

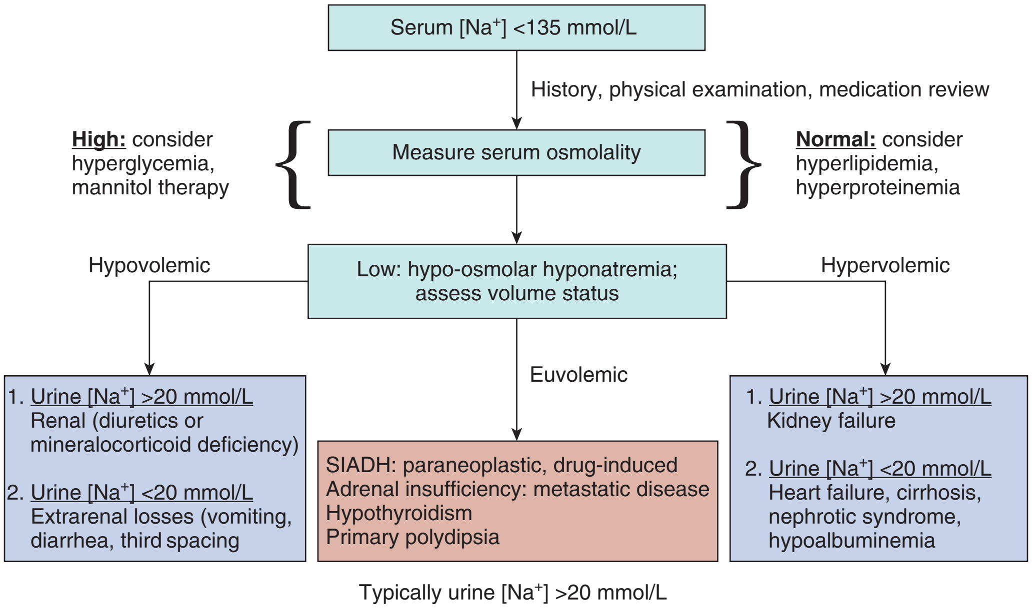

Fig. 42.3 Diagnostic approach to hyponatremia in the patient with cancer.

can be extremely challenging. In these cases, as well as those cases of SIADH in which the urine osmolality is high (>600 mOsm/kg), the use of vasopressin type 2 receptor antagonists (e.g., conivaptan [IV only] or tolvaptan [oral only]) can be considered.[131] These drugs block the action of antidiuretic hormone (ADH) in the distal tubule and allow for the excretion of more dilute urine with increased free water excretion.[131] Tolvaptan is contraindicated in patients with hypovolemic hyponatremia, and volume depletion and in patients who cannot perceive or respond to thirst. In addition, given concerns for hepatoxicity with tolvaptan, the current recommendation is to limit its use to 1 month or less. Finally, oral urea administration can also induce a free water diuresis and can be considered as a therapy for SIADH that is not easily responsive to fluid restriction.[130]

HYPOKALEMIA

Hypokalemia is commonly encountered in patients with cancer.[132] The causes of hypokalemia are often overlapping and multifactorial in a given patient (Table 42.8). Unique causes in the patient with cancer include the following: (1) tubular damage associated with chemotherapeutic agents and antimicrobials (e.g., cisplatin, ifosfamide, amphotericin B, aminoglycosides); (2) tubular damage associated with a light-chain–induced proximal tubulopathy, such as in multiple myeloma; (3) ectopic adrenocorticotropic hormone (ACTH) syndrome; (4) lysozymuria associated with hematologic malignancies; and (5) chemotherapy-induced gastrointestinal losses (vomiting and diarrhea).

Transcellular shifts that occur postphlebotomy, while the blood is in a sample tube, can lead to spurious hypokalemia, especially when the patient has marked leukocytosis (>100,000/μL) or when the blood is kept at room temperature for prolonged periods.[133] When this is suspected, rapid separation of the plasma and storage of blood at 4°C will usually confirm a normal serum potassium level.

Table 42.8 Causes of Hypokalemia in the Patient With Cancer

Type	Cause
Tumor-related	Lysozymuria with acute leukemia Mineralocorticoid excess syndromes • Primary hyperaldosteronism (adrenal carcinoma) • Renin-producing tumors (extremely rare) • Ectopic adrenocorticotropin syndrome Intracellular shifts (pseudohypokalemia) in the setting of high white blood cell counts
Chemotherapy-related	Chemotherapeutic agents (cisplatin, ifosfamide) Antimicrobials and antifungal agents (aminoglycosides, amphotericin B) Excessive gastrointestinal losses (vomiting, diarrhea)
Other-related	• Hypercalcemia • Hypomagnesemia • Postobstructive diuresis • Use of growth factors

The syndrome of paraneoplastic ACTH production can be encountered in several tumors, with the most common causes being bronchial carcinoid tumors, small cell lung carcinoma, lung adenocarcinoma, thymic cancers, and pancreatic tumors.[134] Patients typically present with signs of severe hypercortisolism—diabetes, bone loss, hyperlipidemia, HTN, and, depending on the length of symptoms, cushingoid habitus—along with severe hypokalemia.[134] The hypokalemia results from excessive cortisol production, which overwhelms the distal tubular mechanism that normally limits mineralocorticoid receptor activation by cortisol (11β-hydroxysteroid dehydrogenase). The excessive cortisol levels essentially create a syndrome similar to that of excessive aldosterone secretion, with resulting HTN and hypokalemia. Diagnosis of this syndrome is through measurement of excessive ACTH levels, hypercortisolism, and localization of the primary tumor with radiographic studies. Although optimal management is through control of the primary tumor, this may not be possible, so drugs that antagonize glucocorticoid synthesis (metyrapone and ketoconazole) are often required.[134] Rarely, adrenalectomy may be required.

A rare form of hypokalemia associated with acute myelogenous leukemia (subtypes M4 and M5) has been reported that is usually associated with other electrolyte and acid–base disorders; this includes hyponatremia, hypocalcemia, hypophosphatemia, hypomagnesemia, and a non–anion gap acidosis.[135,136] The magnitude and breadth of these electrolyte issues suggest a global tubular defect and inappropriate renal potassium losses. This defect has been postulated to be secondary to increased tumor-derived lysozyme secretion and lysozymuria-induced tubular damage.[136]

HYPERKALEMIA

Hyperkalemia in patients with cancer is often seen in association with acute kidney injury, rhabdomyolysis, or tumor lysis syndrome. Less common causes include adrenal insufficiency (secondary to adrenal metastases) and drugs (e.g., heparin, ketoconazole, trimethoprim, and calcineurin inhibitors). An important consideration in patients with marked leukocytosis or thrombocytosis is pseudohyperkalemia, which results from a shift of potassium out of platelets or white blood cells postphlebotomy and after clotting of the sample.[137] Although patients will have hyperkalemia reported on laboratory reports, they will not display clinical or electrocardiographic signs of an elevated potassium level. In patients for whom this may be a concern, it is recommended that plasma samples be used for laboratory measurement.

HYPERCALCEMIA

Hypercalcemia that is directly related to malignancy may occur in as many as 30% of patients with cancer at some point during their course.[138,139] When hypercalcemia occurs in these patients, it portends a very poor prognosis and a limited lifespan.[138,139] The symptoms attributable to hypercalcemia are generally nonspecific (e.g., lethargy, nausea, vomiting, confusion) and are related to both the absolute level of serum calcium and the rapidity at which the disorder develops. Patients presenting with serum calcium levels above 13 mg/dL are generally symptomatic.

There are three mechanisms that describe the pathophysiology in most cases of malignancy-associated hypercalcemia (Fig. 42.4). The most common cause is paraneoplastic production of parathyroid-related peptide (PTHrP).[140] PTHrP is very similar to parathyroid hormone (PTH) and acts to increase bone resorption (increases bone calcium release) via activation of osteoclasts and increase reabsorption of calcium in the renal tubules.[140] However, PTHrP is less likely than PTH to stimulate 1,25-dihydroxyvitamin D production. Therefore, PTHrP does not increase intestinal calcium absorption. Hypercalcemia is accompanied by hypophosphatemia, reduced tubular reabsorption of phosphorus, enhanced tubular reabsorption of calcium, and increased excretion of

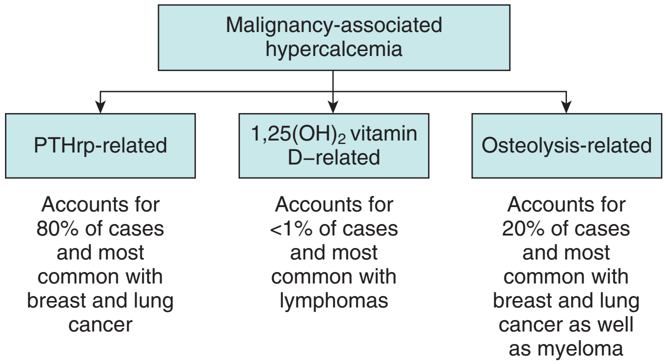

Fig. 42.4 Causes of malignancy-associated hypercalcemia.

nephrogenous cyclic adenosine monophosphate (cAMP), reflecting the PTH-like actions of PTHrP. The second most common mechanism of malignancy-associated hypercalcemia is local osteolytic effects of the cancer and is seen in patients with myeloma, breast cancer, lymphomas, and other cancers. For example, myeloma has a predilection to grow in bone environments, and this is facilitated by the secretion of bone reabsorbing factors such as interleukin-1 (IL-1), IL-6, and tumor necrosis factor, leading to extensive bone destruction, lytic bone lesions, and the release of calcium.[141] Typically, these patients will have advanced cancers, with skeletal surveys demonstrating lytic bone lesions. The third mechanism of malignancy-associated hypercalcemia is due to activation of vitamin D by the tumor itself, which can be seen in lymphomas and, rarely, with myeloma. These patients will have elevated 1,25 dihydroxyvitamin D levels and suppressed PTH levels.[142] The elevated activated vitamin D levels facilitate increase gastrointestinal calcium absorption and increased bone calcium release leading to hypercalcemia.

The diagnostic approach to the patient with hypercalcemia is presented in Fig. 42.5. Once hypercalcemia is confirmed with measurement of an ionized calcium level, the diagnostic algorithm centers on measurement of PTH. PTH levels are suppressed in patients with malignancy-associated hypercalcemia, with the rare exceptions of tumors such as parathyroid adenocarcinomas, which produce excessive PTH. In patients with a suppressed PTH level, measurement of PTHrP and vitamin D levels is the next diagnostic step and allows for accurate diagnosis in most cases.[143]

Patients with hypercalcemia are usually severely volume-depleted due to the effects of elevated calcium levels causing nephrogenic diabetes insipidus, as well as nausea and vomiting. In addition, hypercalcemia, via activation of the calcium-sensing receptor in the thick ascending limb of the loop of Henle, leads to natriuresis and worsening of the volume depletion.[144] Thus, the cornerstone of the treatment of hypercalcemia is aggressive restoration of the extracellular volume, as well as induction of natriuresis with IV fluids (typically, 0.9% saline). This will allow excretion of excess calcium. Loop diuretics should only be added in the presence of volume overload and are not a routine part of management.[145] Data support the use of bisphosphonates as highly effective agents in the management of patients with malignancy-associated hypercalcemia, largely due to their effect to inhibit osteoclast activity.[138] Both pamidronate and zoledronic acid are U.S. Food and Drug Administration

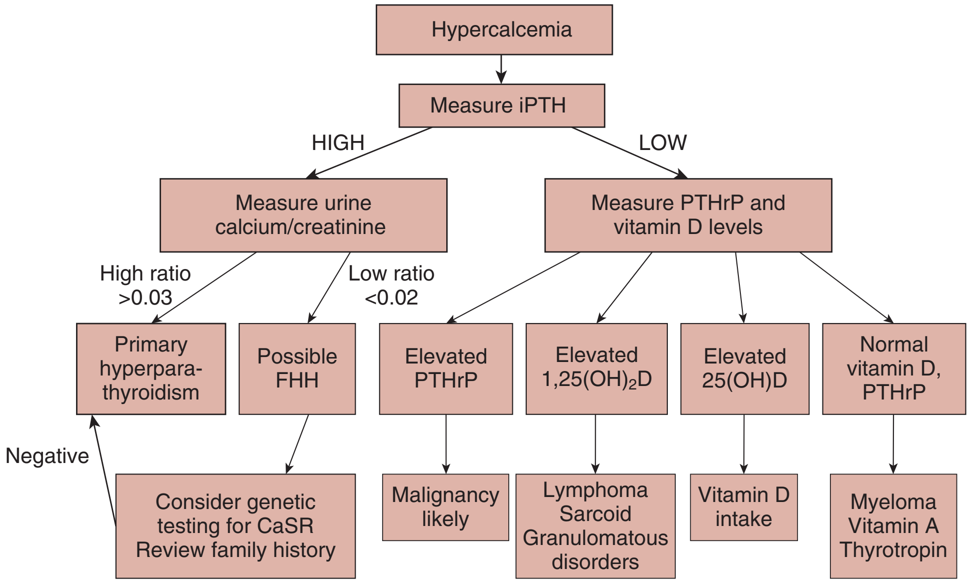

Fig. 42.5 Diagnostic algorithm for the patient presenting with hypercalcemia. Measurement of PTH is the first step in the diagnostic approach. Patients with malignancy-associated hypercalcemia typically have low levels of PTH due to feedback inhibition of the high blood calcium levels on the parathyroid gland. In patients with low levels of PTH, measurement of PTHrP and 1,25-dihydroxyvitamin D (1,25(OH)$_2$D) levels will allow classification of patients into several diagnostic groupings. In patients with elevated PTH levels, a 24-hour urine collection for calcium and creatinine levels should be obtained. *CaSR,* Calcium-sensing receptor; *FHH,* familial hypocalciuric hypercalcemia; *iPTH,* intact parathyroid hormone; *PTH,* parathyroid hormone; *PTHrP,* parathyroid-related peptide; *25(OH)D,* 25-hydroxy vitamin D. (From Reagan P, Pani A, Rosner MH. Approach to diagnosis and treatment of hypercalcemia in a patient with malignancy. *Am J Kidney Dis.* 2013;63(1):141–147.)

Table 42.9 Dosing Guidelines for the Use of Bisphosphonates in the Treatment of Malignancy-Associated Hypercalcemia

Bisphosphonate	Dosing Guidelines	Interval Between Doses
Pamidronate	CrCl > 60 mL/min: 90 mg IV administered over 2–3 hours CrCl, 30–60 mL/min: 60 to 90 mg IV administered over 2–4 hours CrCl < 30 mL/min: 60–90 mg IV administered over 4–6 hours	Every 3–4 weeks
Zoledronate	CrCl > 60 mL/min: 4 mg IV over 15 minutes CrCl, 50–60 mL/min: 3.5 mg IV over 15 minutes CrCl, 40–49 mL/min: 3.3 mg IV over 15 minutes CrCl 30–49 mL/min: 3 mg IV over 15 minutes CrCl < 30 mL/min: not recommended	Every 3–4 weeks

(FDA)–approved for this indication; dosing guidelines are shown in Table 42.9. These agents are generally well tolerated, with the most common side effect being infusion-related fever, but serious nephrotoxicity of these agents has been described, including glomerular lesions and tubular toxicity.[146] Thus, the serum creatinine level should be monitored before each dose of bisphosphonate and the drug withheld in patients with worsening renal function. Urine albumin excretion should be monitored at 3- to 6-month intervals, and the drug should be held if albuminuria develops. Patients with PTHrP-mediated hypercalcemia may remain mildly hypercalcemia, even after bisphosphonate administration, due to the effects of PTHrP to decrease renal calcium excretion. An alternative antiresorptive agent is denosumab, a monoclonal antibody directed against the receptor activator of nuclear factor kappa B ligand (RANKL).[147] Denosumab does not require dose adjustment for renal insufficiency and has demonstrated benefit in reducing skeletal-related events in patients with metastatic cancer.[148] For patients with hypercalcemia secondary to tumors producing 1,25-dihydroxyvitamin D, corticosteroid therapy may be of benefit.[149] Finally, in patients with significant renal failure and hypercalcemia, hemodialysis with a low calcium dialysate can be beneficial.[138]

HYPOPHOSPHATEMIA

Cancer can lead to disorders of phosphate homeostasis via a number of mechanisms. Hypophosphatemia can result in the setting of malnutrition and cachexia and with vitamin D deficiency; these patients typically have low normal serum calcium levels, low vitamin D levels, and elevated PTH levels. In addition, patients with hypercalcemia due to elevated PTH or PTHrP levels may have low-normal serum phosphate levels due to the effects of these hormones on renal phosphorus handling. In some cases, proximal tubular injury (either through chemotherapy such as cisplatin or ifosfamide or due to light-chain–induced tubular injury from myeloma) can lead to phosphaturia and hypophosphatemia.

A rare and unique form of hypophosphatemia is also encountered in patients with cancer, the syndrome of tumor-induced osteomalacia (TIO) or oncogenic osteomalacia.[150] In this syndrome, tumor production of phosphaturic factors such as fibroblast growth factor-23 (FGF-23) results in renal phosphate wasting, hypophosphatemia, and osteomalacia. Tumor types producing these factors include chondrosarcoma, osteoblastoma, and hemangiopericytomas. These tumors may be small and indolent and may require extensive and specialized imaging for localization. Patients with TIO typically have normal serum calcium levels, and the diagnosis is facilitated by measuring the percentage tubular reabsorption of phosphate or the tubular maximum for phosphate, corrected for the GFR.[151] Once renal phosphate wasting is confirmed, measurement of elevated levels of FGF-23 can be diagnostic of TIO. Importantly, 1,25-dihydroxyvitamin D levels are low because FGF-23 inhibits the activity of the 1α-hydroxylase. Therapy rests on surgical removal of the tumor, which is usually curative.

HYPOMAGNESEMIA

Hypomagnesemia is a relatively common electrolyte disorder in patients with cancer; the unique causes in this patient population relate to chemotherapeutic agents such as cisplatin and cetuximab. Otherwise, patients usually develop hypomagnesemia due to gastrointestinal losses, especially with prolonged diarrhea, or renal losses due to diuretic therapy. Other drugs associated with hypomagnesemia in the patient with cancer may include proton pump inhibitors, aminoglycoside antibiotics, calcineurin inhibitors (in patients with stem cell transplants), and amphotericin B.[152]

Cisplatin causes magnesium wasting in more than 50% of treated patients, and the incidence increases with the cumulative dose.[153] Renal magnesium wasting continues after cessation of the drug for several months but can persist for years. The occurrence of magnesium wasting does not correlate with cisplatin-induced acute kidney injury. Cetuximab, a monoclonal antibody against the epithelial growth factor receptor (EGFR), has been associated with hypomagnesemia due to renal losses in as many as 30% of patients receiving this drug.[154] The mechanism of magnesium wasting during treatment with cetuximab is associated with the blockage of the EGFR-dependent transient receptor potential channel 6 (TRPM6) in the nephron, which results in insufficient activation of the TRPM6 epithelial ion channel and magnesium wasting in the distal convoluted tubule. Cessation of therapy with cetuximab leads to reversal of the magnesium wasting and return of the serum magnesium level to normal.

KIDNEY DISEASE ASSOCIATED WITH HEMATOPOIETIC STEM CELL TRANSPLANTATION

HSCT is an important therapy for many cancer patients. However, it is complicated by acute and chronic kidney disease, which are often associated with increased morbidity and mortality. Causes of HSCT-associated kidney injury are often multifactorial, including the conditioning chemotherapy regimen, radiation exposure, nephrotoxic medications, sepsis, sinusoidal obstruction syndrome (SOS), HSCT-associated TMA and GVHD.[155,156] Short- and long-term morbidity and mortality are both increased in the setting of HSCT.[157] Other complications include fluid disturbances, electrolyte abnormalities, glomerular disease, and HTN.

HSCT involves administering lethal doses of chemotherapy and radiotherapy, followed by bone marrow rescue by engraftment of stem or progenitor cells, which are harvested from bone marrow, peripheral blood, or umbilical cord blood. Stem or progenitor cells can be autologous (patient) or allogeneic (donor). In myeloablative HSCT, high-dose chemotherapy and radiation are administered to eradicate the underlying malignancy and bone marrow, followed by marrow reconstitution with the infusion of stem or progenitor cells. However, because myeloablative HSCT is extremely toxic, older patients and those with numerous comorbidities are often excluded from transplantation candidacy. To allow this group of higher risk patients to benefit from HSCT, less toxic nonmyeloablative regimens have been created, using a so-called graft-versus-tumor effect for therapeutic efficacy.

HEMATOPOIETIC STEM CELL TRANSPLANTATION–ASSOCIATED ACUTE KIDNEY INJURY

The incidence of AKI with HSCT is quite varied (range, 15%–73%) due to the AKI definition used, the type of HSCT performed (allogenic vs. autologous), and the chemotherapeutic conditioning regimen used (high dose vs. reduced intensity).[155] The incidence and clinical course of AKI following allogeneic HSCT are well documented. In one study, AKI developed in 53% of patients, of whom half required RRT.[158] Subsequent studies have noted that the incidence of moderate to severe AKI (defined as a doubling of the baseline serum creatinine value) ranges from 36% to 78% with allogenic HSCT.[159–161] Dialysis was required in 21% to 33% of these patients, who suffered a mortality of 78% to 90%. The incidence of AKI with autologous HSCT is less common.[159] In patients with breast cancer, autologous HSCT was associated with the development of moderate to severe AKI in 21% of patients and a mortality rate of 18%. The lower AKI incidence in this group may be due to calcineurin inhibitor avoidance.[162]

Nonmyeloablative HSCT is also associated with a lower incidence of AKI, which is attributed to the milder conditioning regimen used.[163] One study in nonmyeloablative HSCT patients demonstrated a cumulative AKI incidence of 40.4% at 4 months, and dialysis was required in 4.4% of patients. In contrast to myeloablative HSCT, AKI in these patients was usually associated with calcineurin inhibitor exposure. In addition, the development of AKI during nonmyeloablative HSCT occurs over the first 3 months, which is in contrast to myeloablative HSCT, in which AKI typically develops in the first 3 weeks.

As with AKI incidence, the risk for requiring RRT after HSCT varies widely. Overall, approximately 5% of HSCT patients with AKI require dialysis, which is associated with increased mortality risk.[163–169] However, the need for dialysis ranges from 0% to 30%,[9–15] and the risk is higher with myeloablative therapy.[170] Outcomes in patients requiring acute RRT are difficult to quantify accurately. Most

Table 42.10 Risk Factors and Causes of Acute and Chronic Kidney Lesions With HSCT

Acute Kidney Injury	Chronic Kidney Disease
Prerenal—nausea, vomiting, and diarrhea associated with acute gastrointestinal GVHD, drug-induced nausea and vomiting	Thrombotic microangiopathy—calcineurin inhibitors, chronic GVHD
Prerenal/acute tubular injury—sepsis, sinusoidal obstruction syndrome, acute GVHD, marrow infusion syndrome	Arteriolonephrosclerosis—hypertension, radiation, TBI
Thrombotic microangiopathy—acute GVHD, calcineurin inhibitors, TBI	Viral nephropathy—BK virus, cytomegalovirus
Acute tubular injury, crystalline nephropathy—amphotericin, vancomycin, aminoglycosides, acyclovir, other nephrotoxins	Membranous nephropathy, minimal change disease—chronic GVHD, drug-induced glomerular injury

GVHD, Graft-versus-host disease; *HSCT,* hematopoietic stem cell transplantation; *TBI,* total body irradiation.

studies have reported that acute dialysis in this group is associated with an extremely high mortality rate, ranging from 80% to 100%.[156,163–166,168,171,172]

Common risk factors for and causes of AKI after HSCT include volume depletion, sepsis, nephrotoxic medication exposure, SOS, and GVHD (Table 42.10).[156,169,173,174] Because of a propensity for increased gastrointestinal fluid losses and poor oral intake, HSCT patients are highly susceptible to volume depletion and prerenal AKI. GVHD is unique to HSCT and causes tissue and endothelial damage via T-cell and cytokine-mediated injury.[156] The gastrointestinal mucosa is a common site of injury by GVHD, contributing to inadequate fluid intake and increased fluid losses. SOS is also an independent risk factor for AKI and results in a picture similar to hepatorenal syndrome, with sodium avidity and edema formation. Medications commonly prescribed that cause AKI include vancomycin, aminoglycosides, acyclovir, and amphotericin. These agents cause AKI by direct tubular toxicity, acute interstitial nephritis, and crystal-related tubular injury (acyclovir). Calcineurin inhibitors can lead to renal arteriolar vasoconstriction and have been associated with the development of TMA.

MARROW INFUSION SYNDROME

AKI associated with symptoms of nausea, vomiting, and abdominal pain may develop within 24 to 48 hours of HSCT from hemoglobin pigment nephropathy caused by the infusion of lysed red blood cells. The process of stem cell preservation is associated with exposure to various cryoprecipitants, such as dimethyl sulfoxide, which can cause red cell lysis, with hemoglobinuria in the recipient resulting in the precipitation of heme proteins in the distal tubule.[175] Renal vasoconstriction, direct cytotoxicity of hemoglobin, and intratubular hemoglobin cast formation are the mechanisms whereby AKI develops with hemoglobinuria. Management of this now rare complication includes alkalinization of the urine to increases heme solubility and mannitol-induced diuresis to prevent intratubular heme trapping.[176,177]

HEPATIC SINUSOIDAL OBSTRUCTION SYNDROME

AKI following HSCT can also be due to acute liver injury. Hepatic SOS, formerly known as venoocclusive disease, is a commonly encountered complication after HSCT and has been implicated as an independent risk factor for the development of AKI.[175] It typically occurs within 30 days of HSCT, and the incidence varies widely, ranging from 0% to 62.3% (mean, 13.7%).[175] Although the pathophysiology of AKI in the SOS setting is poorly defined, it appears to be a variant of hepatorenal syndrome (hemodynamic AKI). Hepatic sinusoidal obstruction occurs due to iatrogenic damage to sinusoidal endothelial cells and hepatocytes, which results in fibrous narrowing of small hepatic venules and sinusoids. Injury appears to be triggered by the pretransplantation cytoreductive regimen and is more common after allogeneic than autologous HSCT. Furthermore, cytokine release and glutathione depletion in this setting cause hepatocellular necrosis and fibrosis.[178,179] The development of SOS is usually associated with pretreatment with busulfan, cyclophosphamide, and/or total body irradiation.[180]

Criteria have been established for the diagnosis of hepatic SOS. The Seattle criteria require the presence of at least two of three manifestations—jaundice, painful hepatomegaly, and fluid retention or weight gain within 20 days of HSCT.[179,180] The Baltimore criteria require a bilirubin level of >2 mg/dL, plus two or more of the following: painful hepatomegaly, weight gain >5%, or ascites, within 21 days of HSCT.[181] Hepatic SOS severity is graded as mild (the illness fulfills SOS diagnostic criteria but is self-limiting), moderate (SOS subsides with diuretic or analgesic treatment), and severe (SOS persists for more than 100 days, or death).[179]

Hepatic SOS is characterized by painful hepatomegaly, jaundice, fluid retention, and ascites. Patients have hyperdynamic vital signs along with hyponatremia, oliguria, and low fractional excretion of sodium. Urinalysis reveals minimal proteinuria and muddy brown granular casts as a result of the toxic tubular effects of bile acids and bilirubin in the urine. Hypervolemia is usually resistant to diuretics, and spontaneous recovery is rare.

Risk factors that predispose to the development of AKI include a baseline serum creatinine level > 0.7 mg/dL, weight gain, hyperbilirubinemia, and exposure to amphotericin B, vancomycin, or acyclovir. The development of AKI adversely affects survival. In patients who require RRT, the mortality rate approaches 80%.

The management of hepatic SOS has included a number of agents. Small trials using infusions of prostaglandin E, pentoxifylline, and low-dose heparin have shown promise in the prevention and treatment of SOS.[182–184] Smaller trials with defibrotide, an antithrombotic and fibrinolytic agent, have also shown benefit in patients with SOS, especially when used within the first 2 days of HSCT.[185,186] Beneficial effects are thought to be due to the prevention of platelet aggregation and antiinflammatory properties. However, bleeding risk is a concern and must be closely monitored.

ACUTE THROMBOTIC MICROANGIOPATHY

Acute TMA is characterized by endothelial damage that leads to thickened glomerular and arteriolar vessels, fibrin thrombi in capillary loops and arterioles (Fig. 42.6), the presence of fragmented red blood cells, thrombosis, and endothelial cell swelling.[187,188] In the setting of HSCT, patients may develop subclinical kidney disease, AKI, or CKD.[189,190] Given the challenges of obtaining kidney tissue in patients with HSCT, two consensus guidelines have been published outlining clinical criteria for the diagnosis of TMA.[191,192] To meet criteria, the guidelines require the following: (1) schistocytes on a peripheral smear, and (2) an elevated lactate dehydrogenase concentration. The Blood and Marrow Transplant (BMT) Clinical Trials Network also includes AKI (doubling of serum creatinine level), unexplained central nervous system dysfunction, and a negative Coombs test result as part of diagnostic criteria.[191] The International Guidelines from the European Group for Blood and Marrow Transplantation include thrombocytopenia, anemia, and a decreased haptoglobin concentration.[192] Despite these criteria, making a diagnosis of TMA remains challenging and often requires a high index of suspicion, as supported by validation studies and autopsy studies in which clinical criteria often do not correlate with histologic findings.[193]

It is debated whether TMA is a distinct complication of HSCT or a manifestation of several other post-HSCT complications, such as GVHD, infection, and drug toxicity.[194–197] Despite histologic similarities, endothelial cell damage, primarily in the renal vasculature, underlies the pathogenesis of TMA post-HSCT, which differs from thrombotic thrombocytopenic purpura, which is characterized by low levels of von Willebrand factor–cleaving protease ADAMTS13 (genetic or acquired).[198–202] At times, HSCT may unmask a previously undiagnosed genetic mutation in the alternate complement pathway, resulting in atypical hemolytic-uremic syndrome,[203] which may respond to eculizumab.[204]

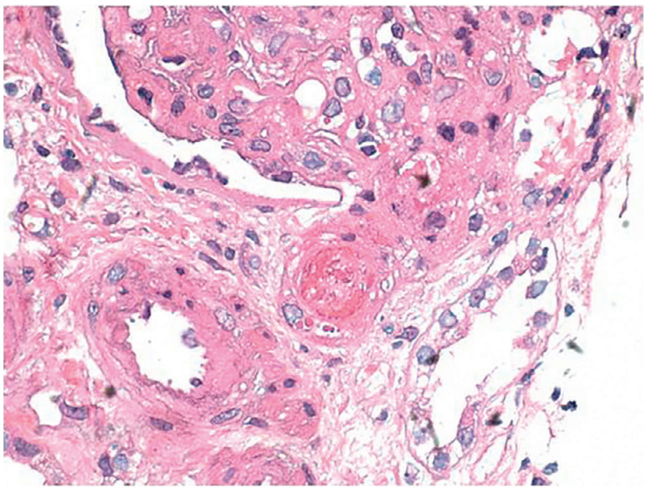

Fig. 42.6 Kidney biopsy from a patient with thrombotic microangiopathy in the setting of HSCT exhibits a fresh fibrin thrombus in a preglomerular arteriole. (Hematoxylin & eosin, ×600.)

In addition to the conditioning regimens for and other exposures during HSCT, some studies have suggested that renal endothelial injury with subsequent TMA may also reflect direct endothelial cell injury by GVHD.[205,206] In fact, the risk of TMA diagnosed on renal pathology at autopsy was fourfold higher in patients with acute GVHD after HSCT. Renal endothelial injury may occur due to circulating inflammatory cytokines or may reflect direct injury to renal endothelial cells by GVHD. Furthermore, plasma markers of endothelial injury and coagulation activation are elevated in patients with acute GVHD after HSCT, suggesting an association among endothelial injury, acute GVHD, and the subsequent development of TMA.[205,206]

Treatment remains challenging for clinicians. Currently available studies are limited by retrospective study designs and the inclusion of heterogeneous patient populations. Options that are used include adjustment of GVHD prophylaxis (either stopping or reducing the calcineurin inhibitor dose), therapeutic plasma exchange (TPE), rituximab, and defibrotide.[191,193,207–209] In uncontrolled studies, reported response rates for TPE have ranged between 27% and 80%.[210–216] In a prospective study of 112 HSCT patients, TMA developed in 11, and 64% of these patients responded with treatment consisting of TPE and cyclosporine withdrawal.[212] Case reports have described a benefit with rituximab treatment of TMA post-HSCT.[192,217–220] More data are required to understand the true utility of rituximab for treatment of this form of TMA.

GRAFT-VERSUS-HOST DISEASE

The use of donor stem cells requires that all allogeneic HSCT recipients receive prophylaxis against GVHD. Prophylactic regimens include cyclosporine A, mycophenolate mofetil, tacrolimus, and sometimes short-term methotrexate.[221,222] GVHD can be classified into acute and chronic categories based on the duration of onset after HSCT and the associated clinical findings. Acute GVHD has great significance because it can independently serve as a risk factor for the development of AKI in HSCT recipients.[223] GVHD may lead directly to AKI through cytokine-mediated inflammation affecting the glomeruli and/or renal tubules or indirectly from exposure to certain medications, such as cyclosporine, which itself can predispose to nephrotoxicity. Furthermore, GVHD also promotes viral reactivation, such as cytomegalovirus, which can contribute to AKI. Gastrointestinal involvement by GVHD is associated with extrarenal fluid losses from vomiting and diarrhea, which can cause prerenal AKI. Treatment options for GVHD include prednisone, antithymocyte globulin, sirolimus, and mycophenolate mofetil.[224] Supportive measures such as IV fluids and avoiding other nephrotoxins are also important in management.

CKD also develops in many patients following HSCT. In can be due to chronic TMA, various viral infections, or a sequelae of acute or chronic GVHD, particularly in patients whose pre-HSCT conditioning regimen did not include TBI.[155,225] A retrospective study of 1635 HSCT recipients has noted that neither TBI nor cyclosporine use was associated with the development of CKD (eGFR < 60 mL/min/1.73 m^2) at 1 year following transplantation. Subgroup analyses of patients receiving cyclosporine have supported the theory that GVHD, independent of calcineurin inhibitor therapy, increases the risk for the development of CKD.[155]

HEMATOPOIETIC STEM CELL TRANSPLANTATION–ASSOCIATED CHRONIC KIDNEY DISEASE

HSCT also appears to be associated with the development of CKD. However, studies reporting the risk of CKD after HSCT are difficult to compare because the definitions used are not consistent, the populations studied are often heterogeneous, and follow-up times vary.[226] In general, the prevalence of CKD after HSCT ranges from 20% to 30%.[227] Microalbuminuria and macroalbuminuria, HTN, and various types of renal tubular dysfunction also complicate the course of HSCT. A meta-analysis has reviewed the published data on CKD after HSCT through 2006.[226] Kidney function was assessed with creatinine-based, radioisotope, or inulin measurements of the GFR. The overall prevalence of CKD was 16.6% (range, 3.6%–89%) and was similar in those receiving autologous and allogeneic transplants, although the allogeneic transplant recipients had a greater decrease in kidney function. Regardless of underlying cause, CKD can progress to ESRD, which increases the mortality risk post-HSCT.[228]

Causes of CKD following HSCT are quite varied (see Table 42.10). The incidence of post-HSCT TMA ranges from 2% to 21%[191] and this complication often follows an indolent course, resulting in CKD and, possibly, progression to ESRD.[229] Although several disease processes, such as TMA, radiation nephritis, nephrotic syndrome, chronic GVHD, and BK virus nephropathy, have been associated with CKD after HSCT, much of the long-term kidney injury in patients undergoing HCT remains unexplained.[230] Multiple bouts of AKI in the setting of HSCT, especially when severe and prolonged, may lead to CKD. Risk factors considered to be potentially important for the development of CKD include acute and chronic TMA post-HSCT, calcineurin inhibitor use (CNI) use, total body irradiation (TBI), and GVHD.[231] CKD occurring from GVHD-associated T-cell and cytokine-mediated damage kidney tissue is thought to be due to a chronic inflammatory state.[232] Glomerular lesions related to HSCT occur primarily in the setting of chronic GVHD. Membranous nephropathy (67%) and minimal change disease (33%) are the two most common pathologies[233,234] and occur within several months of GVHD, approximately 8 to 14 months after HSCT. Hypertension commonly complicates HSCT and likely contributes to CKD and its overall progression. BK virus–related genitourinary and renal lesions also complicate the course of HSCT and are associated with CKD.[235]

ANTICANCER DRUGS AND KIDNEY DISEASE

The number of anticancer agents available to treat various malignancies has grown significantly in the past 10 to 20 years. Chemotherapy-related nephrotoxicity has always been an important area for the nephrologist; however, the number of agents causing kidney injury and an assortment of electrolyte and acid–base disturbances has increased dramatically.[236–242]

Furthermore, understanding renal metabolism and excretion of these drugs is critical in ensuring efficacy while also avoiding or at least reducing toxicity, especially in the setting of acute or chronic kidney disease. Nephrotoxicity associated with the more common chemotherapeutic drugs includes AKI primarily from acute tubular injury, acute interstitial nephritis, and a variety of glomerular injuries; a number of tubulopathies (alone or combined with kidney injury) may develop and result in metabolic perturbations.[236–238] In addition to conventional agents, the emergence of targeted therapies and immunotherapies has further increased the amount of drug-induced nephrotoxicity that is occurring in cancer patients[240–246] (Table 42.11).

Table 42.11 Anticancer Drug Nephrotoxicity

Drug	Clinical Kidney Syndrome	Kidney Histopathology
Conventional Drugs		
Platinum compounds (cisplatin, carboplatin, oxaliplatin)	• AKI • Hypomagnesemia • NDI • Proximal tubulopathy (Fanconi syndrome) • Salt wasting	• Acute tubular injury
Ifosfamide	• AKI • NDI • Proximal tubulopathy (Fanconi syndrome)	• Acute tubular injury
Methotrexate	• AKI	• Acute tubular injury • Crystalline nephropathy
Gemcitabine	• AKI • Hematuria • Hypertension • Proteinuria	• Thrombotic microangiopathy
Mitomycin C	• AKI • Hematuria • Hypertension • Proteinuria	• Thrombotic microangiopathy

Table 42.11 Anticancer Drug Nephrotoxicity (Cont'd)

Drug	Clinical Kidney Syndrome	Kidney Histopathology
Pemetrexed	• AKI • NDI • Proximal tubulopathy (Fanconi syndrome)	• Acute tubular injury • Interstitial fibrosis
Nitrosoureas	• Chronic kidney disease	• Chronic tubulointerstitial nephritis
Targeted Agents		
Anti-VEGF agents (aflibercept, bevacizumab)	• AKI • Hypertension • Proteinuria	• Thrombotic microangiopathy
Tyrosine kinase inhibitors (axitinib, pazopanib, sorafenib, regorafenib, sunitinib)	• AKI • Hypertension • Proteinuria	• Acute tubulointerstitial nephritis • Acute tubular injury • Focal segmental glomerulosclerosis • Thrombotic microangiopathy
BRAF inhibitors (dabrafenib, vemurafenib)	• AKI • Electrolyte disorders	• Acute tubulointerstitial nephritis • Acute tubular injury
ALK inhibitors (crizotinib)	• AKI • Electrolyte disorders • Renal microcysts	• Acute tubulointerstitial nephritis • Acute tubular injury
EGFR inhibitors (cetuximab, erlotinib, gefitinib, panitumumab)	• Hypomagnesemia (secondary hypokalemia and hypocalcemia)	• None
Imatinib	• AKI • Chronic kidney disease	• Acute tubular injury
Rituximab	• AKI (tumor lysis syndrome)	• Acute tubular injury • Crystalline (uric acid) nephropathy
Immunotherapy Agents		
Interferons (alfa, beta, gamma)	• AKI • Nephrotic-range proteinuria	• Focal segmental glomerulosclerosis • Thrombotic microangiopathy
Interleukin-2	• AKI • Capillary leak syndrome	• None
Chimeric antigen receptor (CAR) T cells	• Capillary leak syndrome with AKI, tumor lysis syndrome • Electrolyte disorders	• None
CTLA-4 inhibitors (ipilimumab)	• AKI • Proteinuria	• Acute tubulointerstitial nephritis • Lupus-like glomerulonephritis • Minimal change disease
PD-1 inhibitors (nivolumab, pembrolizumab)	• AKI	• Acute tubulointerstitial nephritis
Other Drugs		
Bisphosphonates (pamidronate, zoledronate)	• AKI • Nephrotic syndrome	• Acute tubular injury • Focal segmental glomerulosclerosis
Sirolimus	• AKI • Proteinuria	• Focal segmental glomerulosclerosis

AKI, Acute kidney injury; *CTLA-4,* cytotoxic T lymphocyte–associated protein-4; *EGFR,* epidermal growth factor receptor; *NDI,* nephrogenic diabetes insipidus; *PD-1,* programmed death-1; *VEGF,* vascular endothelial growth factor.

CONVENTIONAL CHEMOTHERAPEUTIC AGENTS

GEMCITABINE

Gemcitabine is a cell cycle–specific pyrimidone antagonist used to treat a number of malignancies.[246,247] Following gemcitabine approval, TMA was described as a complication, with an incidence ranging from 0.015% to 2.4%.[11–14,246–249] The risk for TMA is associated with a higher cumulative dose and prior exposure to other chemotherapeutic agents that cause TMA.[246–249] Endothelial injury from direct drug toxicity is the most likely mechanism. Disruption of endothelial cell integrity releases von Willebrand factor (vWF) multimers and plasminogen activator inhibitor, reduces the synthesis of prostacyclin and tissue plasminogen activator, and exposes a denuded endothelial surface to fibrin and platelet, which deposit within the renal microvasculature.[246–249]

Gemcitabine-induced TMA usually develops weeks to months after initiating therapy, and AKI develops in most cases.[246–249] A case series of 29 patients with gemcitabine-associated TMA noted microangiopathic hemolytic anemia (MAHA), thrombocytopenia, and AKI in all patients, whereas new or worsening HTN, proteinuria, and hematuria were noted in 26 of 29 patients.[250] Gemcitabine discontinuation or future avoidance resulted in full ($n = 8$) or partial ($n =$

11) recovery, with CKD noted in three and ESRD in seven patients (25%). Plasma exchange and plasmapheresis had minimal or no effect on renal recovery. Reported mortality was wide-ranging (0%–90%), with TMA-specific death estimated at 15%.[246–250]

CISPLATIN

Cisplatin is an effective and widely used chemotherapeutic agent. However, nephrotoxicity is the most common dose-limiting side effect. This relates to its renal site of elimination—, in particular, transport through the proximal tubule.[236–238] Following drug entry into the tubule cell via the organic cation transporter, direct tubular toxicity occurs in part due to a low chloride environment. In the intracellular compartment, chloride molecules are replaced with water molecules in the *cis* position of cisplatin, forming hydroxyl radicals that injure the neutrophilic binding sites on DNA.[236–238,251,252] Nonoliguric AKI, primarily from renal tubular injury (apoptosis and necrosis), typically develops after approximately 3 to 5 days of drug exposure. Furthermore, tubular injury from high-dose cisplatin is often associated with a so-called tubulopathy, which is characterized by various electrolyte (e.g., hypokalemia, hypomagnesemia, hypocalcemia) and acid–base disturbances (e.g., renal tubular acidosis).[2–4,236–238] Hypomagnesemia, which occurs with prolonged exposure, is a challenging management issue because it can be permanent and require IV supplementation[236–238,251,252]

Although kidney injury is generally reversible, repeated cisplatin doses (>100 mg/m^2) may cause permanent kidney injury (GFR reduction of 15%). Isotonic or hypertonic saline administration and avoidance of concomitant nephrotoxins are the most effective ways to prevent cisplatin-induced nephrotoxicity. Amifostine may reduce cisplatin nephrotoxicity through the promotion of improved DNA repair and elimination of free radicals.[2,3,18,236–238,253] Newer platinum agents (carboplatin and oxaliplatin) are less nephrotoxic than cisplatin and are available for patients with underlying kidney disease and for those who previously suffered from cisplatin nephrotoxicity.[254–256] Clinicians must recognize that these agents can still cause AKI and electrolyte disorders.

IFOSFAMIDE

Ifosfamide is an effective alkylating agent that is associated with nephrotoxicity, predominantly injuring the renal tubules. Like cisplatin, this drug enters proximal tubular cells via the organic cation transporter.[236–238] Once inside the cell, the drug or its metabolite chloroacetaldehyde causes direct tubular epithelial cell damage.[236–238] Ifosfamide-induced tubular injury is associated clinically with AKI and a tubulopathy, which can be associated with severe hypokalemia, hypophosphatemia, hypomagnesemia, and hyperchloremia. Patients can experience Fanconi syndrome, with hypophosphatemic rickets and osteomalacia, as well as nephrogenic diabetes insipidus.[236–238,257]

Risk factors for ifosfamide nephrotoxicity include previous exposure to cisplatin, CKD, and a high cumulative dose (>84 g/m^2).[236–238,258,259] Amifostine may offer some protection, whereas mesna, which prevents hemorrhagic cystitis, is ineffective in the kidney. Most patients recover from ifosfamide-induced tubular injury; however, long-term complications such as CKD and tubulopathies may develop. Chronic renal fibrosis, with a progressive decline in the GFR, may rarely lead to ESRD.[236–238,260]

METHOTREXATE

Methotrexate (MTX) is an antifolate drug that is an effective antineoplastic agent when administered in a high dose (>1 g/m^2).[236–238,261–265] Nephrotoxicity occurs primarily because of the precipitation of an insoluble parent drug and metabolites (7-hydroxymethotrexate) within the tubular lumens, a phenomenon known as "crystalline nephropathy."[236–238,261–265] True or effective volume depletion and acidic urine are two major risk factors for the development of AKI; direct tubular toxicity may also contribute to the development of kidney injury. The overall incidence rate of AKI is approximately 1.8% (range, 0%–12%), but can be as high as 50% with high doses in at-risk patients.[236–238,261–265] In general, kidney injury is reversible.

Preventive measures include volume repletion to achieve high urine flow rates. Isotonic saline infusion and furosemide may be necessary to maintain a high urine output. An increase in the clearance rate of MTX (and reduction in crystal precipitation) is seen when the urine pH is increased from 5.5 to 8.4, which can be accomplished with an isotonic solution containing bicarbonate. Once AKI develops, the excretion of MTX is reduced, and systemic toxicity is increased. In this case, therapy is primarily supportive. However, it may be necessary to remove the drug with hemodialysis. Hemodialysis, using high blood flow rates with a high-flux dialyzer, is an effective method of removing MTX.[266] High-dose leucovorin therapy can reduce the systemic toxicity associated with MTX and AKI. Glucarpidase (carboxypeptidase G) may be considered when severe systemic toxicity and AKI are present, and dialysis access is risky. This costly drug metabolizes MTX to benign metabolites.[267]

IMMUNOTHERAPIES

The immune system represents a potent tool to destroy cancer cells. Enhancing the immune system as a therapy for cancer has been used for several decades. Initial therapies included interferon and IL-2. More recently, immune checkpoint inhibitors and chimeric antigen receptor T cells have been used to treat certain forms of malignancy.

INTERFERON

Interferon is used to treat various malignancies (e.g., chronic myelogenous leukemia [CML], renal cell carcinoma) and infectious agents (e.g., hepatitis C). Treatment with interferon results in two renal lesions: (1) podocytopathy with minimal change disease (MCD) and focal segmental glomerulosclerosis (FSGS); and (2) thrombotic microangiopathy.[268,269] MCD was first described but more recent reports have described collapsing and noncollapsing FSGS. Nephrotic-range proteinuria and AKI are observed within weeks to months of commencing interferon therapy. Although proteinuria declines with interferon discontinuation, complete resolution is more common with MCD than with FSGS.[268,269] In particular, collapsing FSGS is associated with CKD. Steroids are associated with complete remission in less than one-third of patients with FSGS. Interferon-related glomerular injury may be due to the direct binding of interferon to podocyte receptors and an alteration of normal cellular proliferation. Macrophage activation and skewing of the cytokine profile toward IL-6 and IL-13, which are possible permeability factors in MCD and FSGS, are also possible mechanisms.[268,269]

INTERLEUKIN-2

High-dose IL-2 has been used to treat renal cell carcinoma, malignant melanoma, and other cancers. High-dose IL-2 is associated with a cytokine release syndrome and capillary leak; this causes a prerenal form of AKI, which develops within 24 to 48 hours.[270] This hemodynamic disturbance is further exacerbated by concurrent NSAID therapy, which is used to reduce fever and myalgias associated with IL-2.[270] Kidney dysfunction generally reverses with drug discontinuation, although ischemic/nephrotoxic acute tubular injury can be associated with a more prolonged course of AKI.[270] This is particularly the case for patients with underlying risk factors, such as CKD, diabetes mellitus, and HTN. To reduce AKI during IL-2 therapy, antihypertensive agents should be withdrawn and fluid resuscitation initiated. Low-dose dopamine (2 μg/kg/min) may prevent some of the renal toxicity of IL-2, reversing oliguria and improving renal recovery time.[271]

IMMUNE CHECKPOINT INHIBITORS

Another method to activate the immune system against the tumor microenvironment is the strategy of immune checkpoint inhibitor (ICI) therapy.[244,245] Activation by an immunologic event such as infection signals cytotoxic T cells to undergo damping of their response to maintain immune homeostasis by expressing several receptors, including cytotoxic T lymphocyte–associated antigen 4 (CTLA-4), programmed death-1 protein (PD-1), and programmed death ligand-1 (PD-L1) that downregulate T-cell function.[244,245,272] These checkpoints normally prevent the development of unwanted autoimmunity. Tumor survival is enhanced in malignancies that overexpress ligands that bind inhibitory T-cell receptors (PD-1), thereby decreasing activated T-cell infiltration into the tumor microenvironment and inhibiting antitumor T-cell responses. To combat this, monoclonal antibody drugs that block ligand binding to PD-1 and CTLA-4 receptors were designed to facilitate T-cell rescue and restore antitumor immunity. Ipilimumab, a fully human, immunoglobulin G1 (IgG1) monoclonal antibody blocking CTLA-4, nivolumab, a fully human IgG4 antibody blocking the PD-1 receptor, and pembrolizumab, a humanized monoclonal IgG4-kappa isotype antibody against PD-1, are the ICIs available in clinical practice. However, blocking immune checkpoints risks the development of pathologic autoimmunity and end-organ injury, such as acute tubulointerstitial nephritis (ATIN). Numerous cases of ATIN, some with granulomatous changes, have been described with the use of ICIs.[244,245] Although some patients develop rash and eosinophilia, AKI is the only consistent clinical manifestation. A small number of cases have required RRT, but most respond to drug discontinuation and steroid therapy.

TARGETED THERAPIES

ANTIANGIOGENESIS DRUGS

The discovery that VEGF is an important mediator of tumor growth and angiogenesis led to the development of drugs targeting this pathway for cancer therapy. However, VEGF is also an essential growth factor for maintaining glomerular and microvascular endothelial health.[240,273] In the glomerulus, VEGF is secreted by podocytes and traverses the basement membrane, where it binds to VEGF receptors on endothelial cells and maintains glomerular endothelial cell integrity and filtration barrier function.[240,273] Anti-VEGF drugs, such as the monoclonal antibody, bevacizumab, as well as drugs that inhibit tyrosine kinase, such as sunitinib, block VEGF function. This leads to glomerular endothelial cell dysfunction and filtration barrier disruption, resulting in proteinuria, thrombotic microangiopathy, HTN, and AKI. Mild and asymptomatic proteinuria is common, although heavy proteinuria occurs in <10% of subjects.[240,273] HTN or aggravation of preexisting HTN is common. The reported incidence varies between 17% and 80%. The effects of anti-VEGF therapy on the development of HTN have been reviewed in a meta-analysis of seven trials, which noted that the relative risk of HTN was three- to sevenfold higher, depending on dose.[274] Hypertension reflects the efficacy of VEGF blockade, perhaps explaining why bevacizumab-induced HTN correlates with clinical outcomes. Hypertension is managed with standard antihypertensive medications, although the use of angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) is preferable in patients with proteinuria. Adverse kidney effects with these drugs may require dose reduction or drug discontinuation, especially with TMA.

EPIDERMAL GROWTH FACTOR RECEPTOR INHIBITORS

The EGF pathway has multiple functions in several organs, including the kidneys.[237,238] Several novel agents that target the EGF pathway are used to treat various cancers; these include receptor antagonists, tyrosine kinase inhibitors, and antisense oligodeoxynucleotides. Cetuximab is a monoclonal antibody against the EGF receptor and was first approved for use in metastatic colon cancer. The primary abnormality observed following EGF receptor antagonism is hypomagnesemia due to renal magnesium wasting.[237,238,275] Hypocalcemia and hypokalemia may also develop as a result of the parathyroid and renal effects of hypomagnesemia.[237,238] EGF normally activates the renal magnesium channel (TRP-M6) in the apical membrane of the distal convoluted tubule to stimulate magnesium absorption. The incidence of severe hypomagnesemia is 10% to 15% but is reversible with drug discontinuation.[276] Many patients require IV magnesium repletion to continue the drug.

B-RAF INHIBITORS

Malignant melanoma frequently has a B-RAF V600 mutation that can be effectively targeted by selective B-RAF inhibition by vemurafenib and dabrafenib. As with other effective cancer agents, these drugs have also been observed to cause nephrotoxicity. Kidney dysfunction, defined as a GFR decline at 1 and 3 months of therapy, occurred in 15 of 16 patients, which was complicated by persistent kidney injury after 8 months of follow-up.[277] In addition, eight patients developed AKI (one with ATN on biopsy) following vemurafenib therapy.[277] AKI that was clinically suggestive of AIN was observed in four patients treated with vemurafenib, with three of four patients recovering kidney function after drug discontinuation, although no biopsy data were available. Of 74 patients treated with vemurafenib, about 60% developed AKI, primarily KDIGO stage 1, within 3 months of drug exposure.[241] Biopsy in two patients revealed tubulointerstitial injury. Kidney function recovered within 3 months of B-RAF discontinuation. AKI and metabolic disturbances from B-RAF inhibitors have been described in the FDA Adverse Events Reporting System.[277] Although the mechanism of kidney injury is unknown, these drugs may interfere with the downstream MAPK pathway, increasing susceptibility to ischemic tubular injury.

ANAPLASTIC LYMPHOMA KINASE INHIBITORS

Anaplastic lymphoma kinase 1 (ALK-1) is a member of the insulin receptor tyrosine kinase family. Small-molecule inhibitors of ALK-1 include crizotinib and ceritinib.[243,278] Crizotinib is an effective agent for the treatment of advanced ALK-positive non–small cell lung cancer; however, it is complicated by two major adverse renal effects. These include AKI and an increased risk for the development and progression of renal cysts.[243,278] In addition to these complications, crizotinib is also associated with the development of peripheral edema and electrolyte disorders (relatively rare). In regard to AKI, it appears that crizotinib is associated with both true AKI (tubulointerstitial injury) and pseudo-AKI, which is likely due to a drug-induced reduction in renal tubular creatinine secretion.

MISCELLANEOUS AGENTS

BISPHOSPHONATES

Bisphosphonates are administered primarily to treat patients with the hypercalcemia of malignancy and to minimize skeletal complications in the setting of bone metastases and multiple myeloma. These agents undergo renal excretion and have been associated with two major types of kidney disease—AKI from acute tubular injury and the podocytopathies, MCD and FSGS.[279] Biopsy-proven ATN has been observed in patients treated with zoledronate, which is often associated with various levels of CKD, despite drug discontinuation.[279] Pamidronate has also been shown to cause AKI, as well as the nephrotic syndrome, from a collapsing variant of FSGS.[280] The mechanism of bisphosphonate-induced kidney injury is not known, but direct toxicity to the glomerular and tubular epithelial cells is suspected. Ibandronate is a newer agent that appears to be significantly less nephrotoxic.[279] Dose modifications are recommended for pamidronate and zoledronate. Pamidronate, 90 mg, administered over 4 to 6 hours (rather than over 2 hours), is recommended for patients with stage 4 CKD, although 60 mg might be a more appropriate dose.[279] The dose of zoledronate should be reduced in patients with stage 3 CKD and avoided in those whose eGFR is <30 mL/min.[279]

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. A 66-year-old man with underlying stage 3b chronic kidney disease (CKD), hypertension, type 2 diabetes mellitus, benign prostatic hypertrophy, and gout undergoes partial nephrectomy for a 3-cm renal cell carcinoma in the left kidney. The operative procedure is associated with a longer than usual warm ischemia time due to complex renal anatomy. Which of the following are associated with the development of progressive CKD in this patient?
   a. Partial nephrectomy
   b. Stage 3b CKD
   c. Type 2 diabetes mellitus
   d. Prolonged intraoperative ischemia time
   e. All of the above

   Answer: e

   Rationale: Patients undergoing nephrectomy for renal cell cancer may develop AKI, CKD, or progressive worsening of underlying CKD. Total (radical) nephrectomy and partial (nephron-sparing surgery) nephrectomy differ in terms of the incidence of postoperative CKD, which is lower with partial nephrectomy due to the reduced nephron loss. However, patients with various medical risk factors for CKD are at increased risk for progressive kidney function loss (potentially progressing to ESRD), irrespective of the use of partial nephrectomy, highlighting the key role played by comorbidities (underlying CKD, hypertension, diabetes mellitus) in the development of postnephrectomy CKD. The duration of intraoperative ischemia is another important predictor of postoperative kidney function. A prolonged warm ischemia time is significantly associated with adverse postoperative kidney function, including both AKI and worsening of CKD.

2. A 42-year-old woman undergoes hematopoietic stem cell transplantation (HSCT) for multiple myeloma. Over the next several weeks, the patient develops a progressive rise in serum creatinine concentration from 1.2 mg/dL to 4.1 mg/dL. A complete blood cell count demonstrates a hemoglobin of 6.8 g/dL and a platelet count of 38. Review of the peripheral blood smear shows a reduced number of platelets and the presence of schistocytes. Concern for the possibility of transplant-associated thrombotic microangiopathy is raised, and laboratory tests show an elevated lactate dehydrogenase level and an undetectable haptoglobin level, with a negative Coombs test result. In addition to lowering the calcineurin inhibitor dose, which of the following is the next best step in the treatment of this patient?
   a. Administration of rituximab
   b. Therapeutic plasma exchange
   c. Red blood cell and platelet transfusion
   d. Administration of IV immunoglobulin (IVIg)
   e. All of the above

   Answer: b

   Rationale: The patient has developed HSCT-associated thrombotic microangiopathy (TMA). Calcineurin inhibitor therapy, used for graft-versus-host disease prophylaxis, has been associated with TMA in the HSCT population. First-line therapy includes lowering of the calcineurin inhibitor use (CNI) dose (or discontinuing the medication) and therapeutic plasma exchange. Most patients respond to these two interventions. Other therapies being studied include rituximab, defibrotide, and eculizumab. However, they are currently not recommended.

3. A 58-year-old man presents with the nephrotic syndrome, normal blood pressure, and a serum creatinine of 1.5 mg/dL, with a benign urinary sediment. A serum protein electrophoresis is normal. The kappa/lambda light-chain ratio is 0.09 (normal, 0.26–1.65); other serologies, including serum complement, are normal. An echocardiogram shows thickening of the septal wall with preserved ejection fraction. The clinical picture is compatible with which diagnosis?
   a. Amyloidosis
   b. Light-chain deposition disease
   c. Both
   d. Neither

   Answer: c

   Rationale: Both immunoglobulin deposition disease and amyloidosis have been associated with cardiac and renal involvement. The presentation at this point is not clearly indicative of one cause over another, and a renal biopsy is required for definitive diagnosis.

4. A 49-year-old female patient has been given a clinical diagnosis of multiple myeloma nephropathy after she presented with progressive renal failure, lytic bone lesions, and thrombocytopenia. Her protein electrophoresis showed high levels of IgG-kappa. Which of the following findings may be present in the kidney biopsy?
   a. Fractured tubular casts
   b. Congo red–positive glomerular and tubular deposits
   c. Acute tubular injury and necrosis
   d. Plasma cell infiltrates
   e. All of the above

   Answer: e

   Rationale: Although cast nephropathy (fractured tubular casts) is the most common finding in myeloma with renal failure, any of the following entities can occur either alone or in combination: amyloidosis (Congo red–positive deposits), light- and heavy-chain deposition disease, fibrillary-immunotactoid glomerulonephritis (GN), Fanconi syndrome-renal tubular acidosis (RTA) from light-chain proximal tubulopathy, membranoproliferative glomerulonephritis (MPGN), infiltration with plasma cells, acute tubular necrosis (ATN), and uric acid nephropathy. Thus, patients with myeloma may benefit from a renal biopsy to provide a definitive diagnosis.